

M13 Batch 5 — Full Set

M13.6 Equipment & Furnishings and M13.8 Instruments · PDF pages 322–448

Cabin equipment and emergency requirements, instrument classification, the atmosphere, pressure measuring devices, pitot static systems and their faults.

332 questions · Level 3: 109 · Level 2: 168 · Level 1: 55

● marks a question built on a fact your instructor highlighted or underlined in the manual (287 of 332). Topics run in manual page order; inside each topic the questions are grouped by level, and the answers with explanations follow at the end of that topic.

No calculations, no unit conversions, no worked examples, nothing type-specific unless it was marked.

Equipment and furnishings: requirements and cabin systems

PDF pages 332–336 · 32 questions

LEVEL 1 — RECALL

1. Which minimum period must a required public address system stay powerable after the shutdown or failure of all engines and auxiliary power units ●
 - a. At least 10 minutes
 - b. At least 3 minutes
 - c. At least 5 minutes
2. Which aggregate period of announcements by flight and cabin crew members is contained within the required powering duration ●
 - a. At least 5 minutes
 - b. At least 10 minutes
 - c. At least 3 minutes
3. On which technology is the Cabin Intercommunication Data System based ●
 - a. A hardwired relay logic
 - b. An analogue amplifier chain
 - c. A microprocessor
4. How many subsystems make up the Advanced Cabin Entertainment/Service System ●
 - a. Four
 - b. Seven
 - c. Five
5. Which subsystem broadcasts flight deck announcements, flight attendant announcements and boarding music to the cabin ●
 - a. The Cabin Interphone System
 - b. The Passenger Address System
 - c. The Passenger Entertainment System

LEVEL 2 — UNDERSTAND AND EXPLAIN

6. A restraining means is fitted to a megaphone. Which paragraph of the design specification sets the inertia forces it must withstand?
 - a. CS 25.1421, Megaphones
 - b. CS 25.1423, Public Address System
 - c. CS 25.561, Emergency Landing Conditions

7. All engines and auxiliary power units have failed. Beyond the announcement time, what further powering does the specification call for? ●
 - a. A standby period for other essential loads on the same source
 - b. A standby period for further announcements by the cabin crew
 - c. A standby period for the crew member interphone system
8. An announcement is made on a required public address system. Where in the aeroplane must it be intelligible? ●
 - a. At all passenger seats, galleys and flight crew seats and work stations
 - b. At all passenger seats, lavatories and cabin crew seats and work stations
 - c. At all passenger seats, lavatories and flight crew rest compartments
9. Which other system must a required public address system be able to work independently of? ●
 - a. Any required passenger entertainment system
 - b. Any required cabin crew member call system
 - c. Any required crew member interphone system
10. Which emergency exits must have a microphone readily accessible to the seated cabin crew member? ●
 - a. Required floor-level exits with an adjacent cabin crew seat
 - b. Required floor-level exits with an adjacent passenger seat
 - c. Required overwing exits with an adjacent cabin crew seat
11. Which three things does the Cabin Intercommunication Data System do to the cabin systems it serves? ●
 - a. Controls, tests and monitors them
 - b. Controls, records and displays them
 - c. Tests, records and announces them
12. Which users are the cabin systems handled by the Cabin Intercommunication Data System dedicated to? ●
 - a. Cabin attendant and passenger use
 - b. Cabin attendant and ground service use
 - c. Flight crew and ground crew use
13. Where does the audio of the Passenger Address System go? ●
 - a. To the passenger headphones and the flight deck speakers
 - b. To the cabin speakers and the passenger headphones
 - c. To the cabin speakers and the attendant handsets
14. Video, announcements and recorded audio all have to reach a seat. Which subsystem controls and distributes them to it? ●
 - a. The Passenger Entertainment System
 - b. The Cabin Interphone System
 - c. The Passenger Services System
15. Smoke is picked up by a lavatory smoke detector. Which cabin subsystem takes that signal and gives the annunciation? ●
 - a. The Cabin Interphone System
 - b. The Cabin Lighting System
 - c. The Passenger Services System
16. The flight deck needs to speak to the load cargo master. Which subsystem carries that call? ●
 - a. The Passenger Address System
 - b. The Cabin Interphone System
 - c. The Cabin Configuration System

17. Which lights are controlled by the Cabin Lighting System ●
 - a. The passenger call lights
 - b. The passenger reading lights
 - c. The ceiling and wash lights
18. Besides the ceiling, wash and night illumination lights, which further lights does the Cabin Lighting System control ●
 - a. The crew rest and reading lights
 - b. The lavatory call and seat row lights
 - c. The lavatory and crew rest lights
19. Which benefit does the Cabin Configuration System give an airline ●
 - a. Fault reporting of the whole cabin to the flight deck
 - b. Interior changes with minimal hardware changes
 - c. Entertainment audio selectable at every seat
20. How is an airplane interior configuration change accomplished through the Cabin Configuration System ●
 - a. By modifying the ACCESS configuration database
 - b. By reprogramming each passenger service unit
 - c. By exchanging the configuration pins of the cabin wiring
21. A new cabin configuration has to be prepared before it is loaded. Which equipment creates the ACCESS configuration database? ●
 - a. An on-board cabin configuration module
 - b. An on-ground personal computer software
 - c. An on-board central maintenance computer
22. An operator wants to alter its speaker volume zones and its smoking areas. Where are those settings held? ●
 - a. In the ACCESS Test System memory
 - b. In the ACCESS configuration database
 - c. In each passenger service unit
23. Which cabin items can be arranged through the ACCESS configuration database ●
 - a. The chime tone and the ceiling light brightness
 - b. The seat recline limits and the galley loads
 - c. The PA areas and the chime mute zones
24. Which test of the ACCESS Test System runs on-line and continuously ●
 - a. The built-in test equipment test
 - b. The built-in test
 - c. The programming test
25. A major fault is found by the ACCESS Test System. Besides the Cabin Configuration/Test Module, where is it reported? ●
 - a. To the passenger service units and the attendant call panels
 - b. To the airplane configuration system and the ground station
 - c. To the EICAS/EFIS interface units and the maintenance computers

LEVEL 3 — APPLY AND REASON

26. The same power source that feeds the public address system also feeds other loads that stay powered when every other source is inoperative. How must that be treated? ●
 - a. Those other loads are shed for the whole powering duration
 - b. Those other loads are taken into account in the powering duration
 - c. Those other loads are transferred to the crew interphone supply
27. A microphone has been left out of its stowage and unused at a cabin crew station. What must the public address system do in that condition? ●
 - a. Alert the flight deck
 - b. Stay serviceable
 - c. Mute the cabin loudspeakers
28. Two required floor-level exits lie close together and each has an adjacent cabin crew member seat. When may one microphone serve both exits? ●
 - a. When the two exits are on the same side of the fuselage
 - b. When the seated crew members can talk unassisted
 - c. When the two crew member seats face in the same direction
29. Two operators want the cabin of the same aeroplane arranged to suit their own service. What does the CIDS design philosophy give them? ●
 - a. A fixed configuration set once by the airframe manufacturer
 - b. Flexibility without complex and time expensive hardware changes
 - c. Flexibility gained by exchanging the cabin wiring looms
30. Which of these systems is a subsystem of the Advanced Cabin Entertainment/Service System ●
 - a. The Cabin Configuration System
 - b. The ACCESS Test System
 - c. The Cabin Lighting System
31. Which two items are the special features of the integrated cabin entertainment and service system rather than its subsystems ●
 - a. The Passenger Services System and the Passenger Address System
 - b. The Cabin Configuration System and the ACCESS Test System
 - c. The Cabin Lighting System and the Cabin Interphone System
32. A reading light and a night illumination light both fail in the same cabin section. Why are two different subsystems involved? ●
 - a. Reading lights belong to the services system and night illumination to the address system
 - b. Reading lights belong to the lighting system and night illumination to the services system
 - c. Reading lights belong to the services system and night illumination to the lighting system

Answers — Equipment and furnishings: requirements and cabin systems

1. a The specification sets a time duration of at least 10 minutes. (L1, PDF p. 332)
2. a An aggregate of at least 5 minutes of crew announcements sits inside the 10 minutes. (L1, PDF p. 332)
3. c The CIDS is described as a microprocessor based system. (L1, PDF p. 334)
4. c Five subsystems are listed for the ACCESS. (L1, PDF p. 336)
5. b Those announcements and the boarding music are broadcast through the Passenger Address System. (L1, PDF p. 336)
6. c The megaphone paragraph itself refers the inertia forces out to CS 25.561. (L2, PDF p. 332)

7. a An additional standby duration is required for the other loads on that source that are essential to safety of flight. (L2, PDF p. 332)
8. b Intelligibility is demanded at the passenger seats, the lavatories and the cabin crew seats and work stations. (L2, PDF p. 332)
9. c Independence is demanded specifically from any required crew member interphone system. (L2, PDF p. 332)
10. a The requirement is written for required floor-level exits that have an adjacent cabin crew member seat. (L2, PDF p. 332)
11. a Control, test and monitor are the three functions printed for the CIDS. (L2, PDF p. 334)
12. a The cabin systems concerned are those dedicated to cabin attendant and passenger use. (L2, PDF p. 334)
13. b PA audio is fed to the cabin speakers and to the passenger headphones. (L2, PDF p. 336)
14. a All three are routed to the seat through the Passenger Entertainment System. (L2, PDF p. 336)
15. c The interface to the lavatory smoke detector for smoke detection annunciation is a feature of the Passenger Services System. (L2, PDF p. 336)
16. b Flight deck to load cargo master is one of the calls made through the Cabin Interphone System. (L2, PDF p. 336)
17. c The ceiling and wash lights are in the Cabin Lighting System list. (L2, PDF p. 336)
18. c The lavatory and crew rest lights complete the Cabin Lighting System list. (L2, PDF p. 336)
19. b The CCS lets the airline change the interior configuration with minimal hardware changes. (L2, PDF p. 336)
20. a Configuration changes are made by modifying the ACCESS configuration database. (L2, PDF p. 336)
21. b The Airplane Configuration System that creates the database is an on-ground PC software. (L2, PDF p. 336)
22. b Speaker volume zones and smoking areas are among the items held in the ACCESS configuration database. (L2, PDF p. 336)
23. c PA areas and chime mute zones are printed in the list of configurable items. (L2, PDF p. 336)
24. b The built-in test is the on-line continuous one, while the BITE is off-line. (L2, PDF p. 336)
25. c Major faults go on to the EIUs and the CMCs as well as to the CCTM. (L2, PDF p. 336)
26. b The duration must be achieved while considering the other loads left on the same source. (L3, PDF p. 332)
27. b No unused, un-stowed microphone may render the system inoperative, so it must remain serviceable. (L3, PDF p. 332)
28. b One microphone may cover several exits only if their proximity allows unassisted verbal communication between the seated cabin crew members. (L3, PDF p. 332)
29. b The CIDS is designed for high flexibility so that complex and time expensive hardware changes are not needed. (L3, PDF p. 334)
30. c The Cabin Lighting System is one of the five subsystems, while the other two are special features. (L3, PDF p. 336)
31. b The CCS and the ATS are named as the two special features added to the five subsystems. (L3, PDF p. 336)
32. c Reading lights sit under the Passenger Services System while night illumination sits under the Cabin Lighting System. (L3, PDF p. 336)

Network integrated cabin equipment

PDF pages 338–338 · 4 questions

LEVEL 1 — RECALL

33. Which kind of system is the nice system ●
- a. A modular IP based cabin management system
 - b. A centralised IP based flight management system
 - c. A modular analogue cabin lighting system

LEVEL 2 — UNDERSTAND AND EXPLAIN

34. For which aircraft is the nice cabin management system offered ●
- a. Regional commuter aircraft
 - b. Long range freighter aircraft
 - c. VIP and government aircraft
35. Which cabin functions can the crew and the passengers reach through the nice system ●
- a. Window-shades, galley power, audio/video and seat heating
 - b. Illumination, cabin pressure, audio/video and internet access
 - c. Illumination, window-shades, audio/video and internet access
36. Further functions are to be added to an existing nice installation. Which protocol makes the system easy to expand? ●
- a. The standard TCP/IP protocol
 - b. The ARINC 429 protocol
 - c. The spanning tree protocol

Answers — Network integrated cabin equipment

33. a The nice system is a modular, IP based cabin management system. (L1, PDF p. 338)
34. c The system is offered for VIP and government aircraft. (L2, PDF p. 338)
35. c Illumination, window-shades, audio/video and internet access are the cabin functions printed. (L2, PDF p. 338)
36. a Expandability comes from the use of the standard TCP/IP protocol. (L2, PDF p. 338)

Instrument classification and cockpit layout

PDF pages 347–361 · 19 questions

LEVEL 1 — RECALL

37. To which two subgroups can every instrument of the classical cockpit be assigned ●
- a. Airframe monitoring and aircraft monitoring
 - b. Flight monitoring and engine monitoring
 - c. Flight monitoring and aircraft monitoring
38. To which group does the artificial horizon indicator ADI belong ●
- a. Flight monitoring
 - b. Airframe monitoring
 - c. Aircraft monitoring
39. A panel layout is described as Basic T. Which four instruments form that T? ●
- a. Attitude indicator, altimeter, compass and turn coordinator
 - b. Airspeed indicator, attitude indicator, altimeter and vertical speed indicator
 - c. Airspeed indicator, attitude indicator, altimeter and compass
40. Where is the attitude indicator located in the Basic T configuration ●
- a. In the middle at the bottom
 - b. On the left at the top
 - c. In the middle at the top

LEVEL 2 — UNDERSTAND AND EXPLAIN

41. The group of instruments which the standby set of a modern cockpit duplicates is ●
- a. airframe monitoring
 - b. flight monitoring
 - c. engine monitoring
42. An exhaust gas temperature indicator is fitted in a classical cockpit. Which group and subdivision does it belong to? ●
- a. Aircraft monitoring, as an engine monitoring device
 - b. Aircraft monitoring, as an airframe monitoring device
 - c. Flight monitoring, as a pressure monitoring device
43. Into which two kinds of devices is aircraft monitoring divided ●
- a. Airframe monitoring devices and engine monitoring devices
 - b. Flight monitoring devices and engine monitoring devices
 - c. Airframe monitoring devices and standby devices
44. Why do the aircraft monitoring systems of the modern cockpit carry different names such as EICAS or ECAM ●
- a. Because the name depends on the display size
 - b. Because the name depends on the flight phase
 - c. Because the name depends on the aircraft type
45. Which task do EICAS and ECAM perform in the modern cockpit ●
- a. They perform maintenance data recording
 - b. They perform flight monitoring
 - c. They perform aircraft monitoring

46. Why is the CMS counted as part of aircraft monitoring ●
- Because it delivers information to the flight crew
 - Because it delivers information to the maintenance personnel
 - Because it delivers information to the standby instruments
47. The electronic displays of a glass cockpit are lost in flight. Which speed and attitude information remains available from the standby set? ●
- Airspeed from the standby airspeed indicator and heading from the standby horizon
 - Mach number from the standby machmeter and attitude from the standby horizon
 - Airspeed from the standby airspeed indicator and attitude from the standby horizon
48. A fuel pressure indication changes during flight. Which value is the instrument responding to?
- Pressure of fluids
 - Earth's magnetism
 - Pressure of air
49. Of the two combined indicators, the one that adds the flight director indication and the one that adds radio navigation indications are ●
- the ADI and the HSI respectively
 - the HSI and the ADI respectively
 - the ADI and the magnetic standby compass respectively
50. A pilot reads his magnetic heading and a VOR course deviation from one round instrument. Which indicator is he using? ●
- The variometer
 - The HSI
 - The ADI
51. A glass cockpit aircraft has no separate round flight instruments. Which statement describes the arrangement of the flight parameters? ●
- They are laid out on the PFD and correspond to the Basic T configuration
 - They are laid out on the PFD in a configuration that replaces the Basic T
 - They are laid out on the navigation display in Basic T configuration

LEVEL 3 — APPLY AND REASON

52. An aircraft presents its altimeter, airspeed indicator and artificial horizon indicator on electronic displays. To which system of the modern cockpit are these instruments assigned? ●
- EICAS (Engine Indicating and Crew Alerting System)
 - EFIS (Electronic Flight Instrument System)
 - CMS (Central Maintenance System)
53. The standby devices of a modern cockpit cover all but one of the flight parameters shown on the PFD. The parameter left without a standby device is ●
- altitude
 - airspeed
 - vertical speed
54. An engineer needs aircraft system information for troubleshooting rather than an indication meant for the crew. Which aircraft monitoring system delivers it? ●
- The ECAM (Electronic Centralized Aircraft Monitoring)
 - The EFIS (Electronic Flight Instrument System)
 - The CMS (Central Maintenance System)

55. An instrument is to be fitted in the middle at the top of a Basic T panel and must also display the indication from the flight director system. Which indicator is required? ●
- a. The attitude director indicator ADI
 - b. The horizontal situation indicator HSI
 - c. The turn coordination indicator

Answers — Instrument classification and cockpit layout

37. c The text splits classical cockpit instruments into flight monitoring and aircraft monitoring. (L1, PDF p. 347)
38. a The ADI is listed among the flight monitoring instruments. (L1, PDF p. 347)
39. c The four instruments arranged in the T shape are the airspeed indicator, attitude indicator, altimeter and compass; turn coordination and vertical speed sit below them. (L1, PDF p. 361)
40. c The Basic T places the attitude indicator centrally at the top of the group. (L1, PDF p. 361)
41. b Altimeter, airspeed indicator, horizon and compass are all listed as flight monitoring instruments. (L2, PDF p. 347)
42. a Exhaust gas temperature indicators are named as engine monitoring devices under aircraft monitoring. (L2, PDF p. 347)
43. a Aircraft monitoring covers airframe monitoring devices and engine monitoring devices. (L2, PDF p. 347)
44. c The text states that aircraft monitoring instruments have different names depending on aircraft type. (L2, PDF p. 347)
45. c EICAS and ECAM are the names given to the aircraft monitoring instruments. (L2, PDF p. 347)
46. b The CMS is used for aircraft monitoring because it delivers information to maintenance personnel. (L2, PDF p. 347)
47. c The standby set contains an airspeed indicator and a horizon, but no machmeter. (L2, PDF p. 347)
48. a Fluid pressure is listed separately from air pressure as a value that operates instruments. (L2, PDF p. 347)
49. a The ADI adds the flight director indication to the artificial horizon; the HSI adds radio navigation indications to the compass. (L2, PDF p. 361)
50. b The HSI is the instrument that combines compass and radio navigation information. (L2, PDF p. 361)
51. a The PFD keeps the Basic T arrangement rather than abandoning it. (L2, PDF p. 361)
52. b All three named instruments are listed as flight monitoring instruments, and flight monitoring instruments go to the EFIS. (L3, PDF p. 347)
53. c The standby set holds altimeter, airspeed indicator, horizon and compass, so of the PFD parameters only vertical speed has no standby device. (L3, PDF p. 347)
54. c The CMS is the aircraft monitoring system that delivers information to the maintenance personnel. (L3, PDF p. 347)
55. a The middle top position of the Basic T carries the attitude indicator, and the ADI is the attitude indicator that also shows the flight director indication. (L3, PDF p. 361)

The atmosphere, ISA and terminology

PDF pages 365–375 · 34 questions

LEVEL 1 — RECALL

56. What is the proportion of nitrogen in dry, clean air directly above the surface of the earth ●
- 78%
 - 21%
 - 1%
57. What is the proportion of oxygen in dry, clean air directly above the surface of the earth ●
- 21%
 - 11%
 - 78%
58. An air data system is being described to a student. Which two measurements does flight monitoring require? ●
- Static pressure and dynamic pressure
 - Dynamic pressure and air temperature
 - Static pressure and air density
59. What is the other name for dynamic pressure ●
- Atmospheric pressure
 - Barometric pressure
 - Ram pressure
60. What does HUD stand for ●
- Heading Up Display
 - Head Unit Display
 - Head Up Display
61. The transition level is the specified last usable flight level. By what minimum margin must it lie above the 5000 ft MSL level ●
- 500 ft
 - 2000 ft
 - 1000 ft

LEVEL 2 — UNDERSTAND AND EXPLAIN

62. How does the temperature behave with increasing height in the troposphere ●
- It decreases by 6.5 °C every 100 metres
 - It increases by 0.65 °C every 100 metres
 - It decreases by 0.65 °C every 100 metres
63. The air data measurement also called atmospheric pressure and the one also called ram pressure are ●
- the static pressure and the total pressure respectively
 - the dynamic pressure and the static pressure respectively
 - the static pressure and the dynamic pressure respectively
64. The two indications that become possible once temperature is applied as a corrective value are ●
- airspeed and vertical speed
 - barometric altitude and true airspeed
 - true airspeed and static air temperature

65. How do static pressure, temperature and density behave as altitude increases ●
- a. Only temperature becomes lower
 - b. All three become lower
 - c. All three become higher
66. True airspeed and static air temperature are to be indicated as well. Which value must be applied as a corrective? ●
- a. Humidity
 - b. Temperature
 - c. Density
67. A pneumatic instrument is calibrated in the workshop. Which values form the basis of that calibration?
- a. The actual airfield values at the time of installation
 - b. The tropopause values of the standard atmosphere
 - c. The standard atmosphere values at mean sea level
68. Why has the international standard atmosphere been established
- a. To set the pressure scale of the altimeter before every flight
 - b. To predict the weather conditions along a planned route
 - c. To compare flight performance under greatly differing conditions
69. In a modern indicating system all the air data information is presented ●
- a. on separate electrical instruments
 - b. on one screen called the HUD
 - c. on one screen called the PFD
70. How does the HUD present the PFD data ●
- a. So that the pilot can see it when he looks through the windshield
 - b. So that the pilot can see it only when the autopilot is engaged
 - c. So that the pilot can see it when he looks down at the centre pedestal
71. The one of the two transition terms that applies prior to landing is ●
- a. the flight level
 - b. the transition level
 - c. the transition altitude
72. A barometric altimeter is set to QFE while the aircraft stands on the airfield. What does it indicate?
- a. Current flight level
 - b. Airfield elevation
 - c. Zero altitude
73. A crew selects QNH on both altimeters. In which phases of flight is this setting used? ●
- a. During cruise, above transition altitude and above transition level
 - b. During landing and take-off, up to transition altitude and below transition level
 - c. During landing and take-off, above transition altitude and above transition level
74. What is the transition altitude ●
- a. Prior to landing, the height at which the altimeter is set from QNH to STD
 - b. After take-off, the height at which the altimeter is set from QNH to STD
 - c. After take-off, the height at which the altimeter is set from STD to QNH

75. What is the transition level ●

- a. Prior to landing, the last usable flight level at which the altimeter is set from STD to QNH
- b. Prior to landing, the last usable flight level at which the altimeter is set from QNH to STD
- c. After take-off, the last usable flight level at which the altimeter is set from STD to QNH

LEVEL 3 — APPLY AND REASON**76. A crew flies at a level that lies inside the troposphere over the equator but above it over the poles. Which printed thicknesses account for this? ●**

- a. About 16.8 km at the equator and about 11 km at the poles
- b. About 8.5 km at the equator and about 16.8 km at the poles
- c. About 16.8 km at the equator and about 8.5 km at the poles

77. An aircraft climbs and its static pressure falls. The indication a mechanical air data instrument derives from that falling pressure is ●

- a. the indicated airspeed
- b. the static air temperature
- c. the barometric altitude

78. An air data system has static and dynamic pressure available but no temperature input. The indication it cannot produce is ●

- a. the true airspeed
- b. the vertical speed
- c. the barometric altitude

79. An airspeed indicator with bellows works on the difference between total and static pressure. Of the two air data measurements required for flight monitoring, the one it uses for this is ●

- a. the dynamic pressure
- b. the static pressure
- c. the barometric altitude

80. In cruise above the transition altitude all pilots set 1013.2 hPa on the altimeter. Which standard atmosphere value is this setting taken from? ●

- a. The pressure at the transition level
- b. The pressure at mean sea level
- c. The pressure at the tropopause

81. How does the air temperature of the standard atmosphere behave with increasing height directly above the tropopause

- a. It continues to fall at the same rate
- b. It stops falling and stays constant
- c. It begins to rise with height

82. An aircraft has electrical air data instruments fed from separate air data calculators. In this fit the conversion from pressure by mechanical means takes place

- a. in the instrument itself
- b. separately from the instrument
- c. in the pitot and static probes

- 83.** One screen in the cockpit carries all the air data information and also every Basic T parameter. This screen is ●
- the HUD (Head Up Display)
 - the PFD (Primary Flight Display)
 - the standby horizon
- 84.** Two altimeters on the same parked aircraft are read at the airfield, one showing zero and the other showing the airfield elevation. Which settings are selected?
- QFE on the first and QNE on the second
 - QFE on the first and QNH on the second
 - QNH on the first and QFE on the second
- 85.** An aircraft climbs through the transition altitude after take-off. Which change of altimeter setting is required? ●
- From QNH to STD (QNE)
 - From STD (QNE) to QNH
 - From QFE to QNH
- 86.** An aircraft descends through the transition level on the approach. Which change of altimeter setting is required? ●
- From QNH to STD (QNE)
 - From STD (QNE) to QNH
 - From QNH to QFF
- 87.** The low range radio altimeter is unserviceable and the crew needs the height above the airfield. Which setting gives it on the barometric altimeter, and under which condition?
- QFE, in as far as the aircraft stays below the transition altitude
 - QNH, in as far as there is no change in air pressure
 - QFE, in as far as there is no change in air pressure
- 88.** An aircraft in cruise above the transition altitude keeps QNH set instead of the standard pressure. What is no longer ensured
- the indication of the airfield elevation
 - the measurement of the height above the ground
 - the safe vertical separation from other aircraft
- 89.** An aircraft cruises at 4000 ft above mean sea level in an area with a transition altitude of 5000 ft. Which altimeter setting must be used?
- QNE
 - QNH
 - QFF

Answers — The atmosphere, ISA and terminology

- 56. a** Nitrogen makes up 78% of dry clean air at the surface. (L1, PDF p. 365)
- 57. a** Oxygen makes up 21% of dry clean air at the surface. (L1, PDF p. 365)
- 58. a** Flight monitoring is built on the two pressures, static and dynamic. (L1, PDF p. 367)
- 59. c** Dynamic pressure is the ram pressure produced by the aircraft's motion. (L1, PDF p. 367)
- 60. c** HUD is short for Head Up Display. (L1, PDF p. 373)
- 61. c** The manual requires at least 1000 ft of margin above the 5000 ft MSL level. (L1, PDF p. 375)
- 62. c** The troposphere lapse rate is 0.65 °C per 100 m of climb. (L2, PDF p. 365)

63. c The manual gives atmospheric pressure as the other name for static pressure and ram pressure as the other name for dynamic pressure. (L2, PDF p. 367)
64. c Temperature as a corrective value makes true airspeed and static air temperature available. (L2, PDF p. 367)
65. b All three listed values fall with increasing altitude. (L2, PDF p. 367)
66. b Temperature is the corrective value that makes TAS and SAT available. (L2, PDF p. 367)
67. c Calibration is referred to the ISA mean sea level values. (L2, PDF p. 367)
68. c The ISA exists so performance figures can be compared on a common basis. (L2, PDF p. 369)
69. c Modern indicating systems gather all the air data information onto the PFD. (L2, PDF p. 373)
70. a The HUD projects the PFD information into the pilot's forward view. (L2, PDF p. 373)
71. b The transition altitude belongs to the climb after take-off and the transition level to the approach prior to landing. (L2, PDF p. 375)
72. c With QFE set the reading starts from the airfield, so it is zero on the ground. (L2, PDF p. 375)
73. b QNH belongs to the low level phases, below the transition altitude or level. (L2, PDF p. 375)
74. b The transition altitude is the climb-out point for changing to the standard setting. (L2, PDF p. 375)
75. a The transition level is the descent point for changing back to QNH. (L2, PDF p. 375)
76. c The troposphere reaches much higher at the equator than at the poles, so the same level can be inside it in one place and above it in the other. (L3, PDF p. 365)
77. c Static pressure falls as altitude increases, and it is the static pressure that a mechanical instrument uses for barometric altitude. (L3, PDF p. 367)
78. a Barometric altitude, vertical speed and airspeed come from the two pressures alone, while true airspeed needs temperature as a corrective value. (L3, PDF p. 367)
79. a Dynamic pressure is the difference between total and static pressure, so the bellows instrument is working on the dynamic pressure of the two measurements. (L3, PDF p. 367)
80. b The standard atmosphere fixes the sea level pressure at 1013,25 hPa, which is the pressure the crew sets for flight levels. (L3, PDF p. 369)
81. b Identical temperatures at both levels show the lapse rate has stopped at the tropopause. (L3, PDF p. 369)
82. b Where the instruments are electrical and the calculators separate, the mechanical conversion happens in those calculators and not in the instrument. (L3, PDF p. 371)
83. b The PFD is the screen that holds all the air data information, and it shows every one of the Basic T parameters. (L3, PDF p. 373)
84. b With QFE the barometric altimeter reads zero on the airfield, while with QNH it reads the elevation there. (L3, PDF p. 375)
85. a The transition altitude is the height at which the altimeter is changed from QNH to the standard setting. (L3, PDF p. 375)
86. b The transition level is the flight level at which the altimeter is changed from the standard setting back to QNH. (L3, PDF p. 375)
87. c Height is normally taken from the low range radio altimeter, and the barometric altimeter on QFE gives it only as long as the air pressure does not change. (L3, PDF p. 375)
88. c The common 1013.2 hPa setting in cruise is what ensures safe vertical separation, so an aircraft left on QNH no longer has it. (L3, PDF p. 375)
89. b Standard setting only applies above the transition altitude, so below it the altimeter stays on QNH. (L3, PDF p. 375)

Pressure measuring devices

PDF pages 377–393 · 75 questions

LEVEL 1 — RECALL

90. An operator prepares its fleet for RVSM operations. What does this abbreviation mean? ●
- a. The halving of vertical safety distances at higher flight altitudes
 - b. The halving of lateral safety distances at higher flight altitudes
 - c. The doubling of vertical safety distances at higher flight altitudes
91. What is RVSM airspace ●
- a. Airspace above FL 290 with 2000 ft vertical separation
 - b. Airspace above FL 285 with 500 ft vertical separation
 - c. Airspace above FL 290 with 1000 ft vertical separation
92. What does the abbreviation IAS stand for ●
- a. Inertial airspeed
 - b. Increased airspeed
 - c. Indicated airspeed
93. Which airspeed is displayed in aircraft equipped with air data computers and glass cockpits ●
- a. The equivalent airspeed
 - b. The calibrated airspeed
 - c. The true airspeed
94. For which purpose is the true airspeed needed ●
- a. For engine trim purposes
 - b. For navigation purposes
 - c. For design data purposes
95. What forms the sensing element of an aneroid barometer ●
- a. A small chamber made of thin glass
 - b. A small tube filled with thin fluid
 - c. A small capsule made of thin metal
96. What is the function of a Bourdon tube ●
- a. It converts vibration into a proportional electrical signal
 - b. It converts pressure into a proportional electrical signal
 - c. It converts pressure into a proportional temperature signal
97. Which cross-sectional shape does the curved tube of a Bourdon transducer have ●
- a. An oval shape
 - b. A round shape
 - c. A square shape
98. Which stroke do diaphragm capsules supply as the measurable value ●
- a. 4 to 5 mm
 - b. 14 to 15 mm
 - c. 0.4 to 0.5 mm

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 99.** What is the purpose of reducing the vertical separation minimum ●
- To allow aircraft to climb above the tropopause
 - To reduce the workload of the air data computers
 - To accommodate more aircraft in the same airspace
- 100.** Why was the separation above FL 285 formerly increased
- Lack of autopilot authority and reduced control response at high altitude
 - Lack of measurement precision and reduced altimeter sensitivity at high altitude
 - Lack of radar coverage and reduced transponder range at high altitude
- 101.** Besides the pitot-static sensors, the air data computers, the altimeters and the autopilot in altitude hold mode, which further item is listed as RVSM-sensitive? ●
- The structure and surroundings of the pitot-static sensors
 - The radio altimeter antennas and their surroundings
 - The transponder antenna and its surroundings
- 102.** What is the function of the warning device required for flight in RVSM airspace ●
- It indicates when the aircraft leaves the predetermined height
 - It indicates when the altimeter setting differs from 1013.2 hPa
 - It indicates when the aircraft enters RVSM airspace
- 103.** Which autopilot capability is required for flight in RVSM airspace ●
- It must be able to maintain the altitude
 - It must be able to maintain the airspeed
 - It must be able to maintain the heading
- 104.** Which transponder requirement applies to an aircraft flying in RVSM airspace? ●
- It must transmit altitude and be connected to the altitude measurement unit
 - It must transmit altitude and be connected to the radio altimeter
 - It must transmit heading and be connected to the altitude measurement unit
- 105.** Which operational advantage is gained by an RVSM approved aircraft ●
- It can use the increased number of flight levels to save time and fuel
 - It can use a reduced number of flight levels to save maintenance costs
 - It can use the increased number of flight levels to fly above the tropopause
- 106.** A workshop is being prepared for the maintenance of RVSM approved aircraft. Which of the mandatory training subjects concerns the accuracy of the workshop's own equipment?
- The configuration of air data systems
 - Improved altimeter precision
 - Improved precision of test equipment
- 107.** A fleet of six aircraft is to be cleared for RVSM. Who must apply for the Airworthiness Approval and for what? ●
- The aircraft manufacturer, for each individual aircraft
 - The aircraft operator, for each individual aircraft
 - The aircraft operator, for the whole fleet at once

- 108.** What does the requirement for RVSM Airworthiness Approval mean for each individual aircraft ●
- Replacement of the air data computers and the altitude monitoring programme
 - Modification according to required service bulletins and a full static system overhaul
 - Modification according to required service bulletins and the altitude monitoring programme
- 109.** The RVSM training covers surface waviness in the critical area. Which causes of that waviness are named? ●
- Fasteners, static dischargers and paint drops
 - Rivets, oxygen discharge discs and corrosion pits
 - Fasteners, oxygen discharge discs and paint drops
- 110.** Which detail of the air data computer must the mandatory RVSM training cover ●
- Its cooling airflow
 - Its part numbers
 - Its software loading procedure
- 111.** Which approval makes the aircraft eligible for the RVSM test flight ●
- The MNPS Approval
 - The RVSM Operational Approval
 - The RVSM Airworthiness Approval
- 112.** The altitude monitoring programme and the test flight have both been completed successfully. Which approval does the operator now receive? ●
- The MNPS Approval
 - The RVSM Operational Approval
 - The RVSM Airworthiness Approval
- 113.** An airspeed indicator with bellows is connected to the pitot and static systems. Which pressure does its indication directly follow? ●
- The total pressure as the sum of dynamic and ambient pressure
 - The dynamic pressure as the difference between total and static pressure
 - The static pressure as the difference between ambient and total pressure
- 114.** Under which conditions does an airspeed indicator with bellows show the real speed of the aircraft ●
- At cruise altitude with standard conditions
 - At mean sea level with standard conditions
 - At mean sea level with compressibility corrected
- 115.** An air data computer converts the indicated airspeed into the calibrated airspeed. Which two errors does it remove? ●
- The error from the indicator and the error from air compressibility
 - The error from the pitot source and the error from air density
 - The error from the indicator and the error from the static pressure source
- 116.** A mechanical standby airspeed indicator with bellows is fed with total and static pressure. Which airspeed does it display? ●
- The indicated airspeed
 - The equivalent airspeed
 - The calibrated airspeed

- 117.** Of the two speeds derived after the calibrated airspeed, the one needed for navigation and the one kept for design data are ●
- a. the true airspeed and the equivalent airspeed respectively
 - b. the equivalent airspeed and the true airspeed respectively
 - c. the true airspeed and the indicated airspeed respectively
- 118.** Under which condition is the speed formula valid ●
- a. Only when the medium used is at standard temperature
 - b. Only when the medium used is incompressible
 - c. Only when the medium used is above the tropopause
- 119.** An aircraft accelerates to a high speed and the total pressure rises more than the speed formula predicts. What produces this additional pressure? ●
- a. Air expands at higher speeds and generates a total pressure decrease
 - b. Air compresses at higher speeds and generates a total pressure increase
 - c. Air compresses at lower speeds and generates a static pressure increase
- 120.** How does the compressibility error change with airspeed and altitude ●
- a. The lower the airspeed and the higher the altitude, the larger the error
 - b. The higher the airspeed and altitude, the larger the error
 - c. The higher the airspeed and altitude, the smaller the error
- 121.** What is the equivalent airspeed ●
- a. The true airspeed after the density calculation has been removed
 - b. The calibrated airspeed after the compressibility error has been removed
 - c. The indicated airspeed after the instrument error has been removed
- 122.** Why is the equivalent airspeed not shown to the crew of a commercial airplane ●
- a. It is only valid below the tropopause in standard conditions
 - b. It is only available when the air data computer is in test
 - c. It is only important for design data and further calculations
- 123.** A navigation computation needs the speed of the aircraft through the surrounding air mass. Which airspeed is this? ●
- a. The equivalent airspeed
 - b. The true airspeed
 - c. The calibrated airspeed
- 124.** In which order are the airspeeds derived from the pressure inputs
- a. IAS, CAS, EAS, TAS
 - b. IAS, EAS, CAS, TAS
 - c. CAS, IAS, TAS, EAS
- 125.** Which calculation is made between EAS and TAS ●
- a. A static source correction
 - b. A density calculation
 - c. A compressibility correction
- 126.** Which correction is applied between CAS and EAS ●
- a. Correction of static source error
 - b. Correction of compressibility error
 - c. Correction of instrument error

127. The compressibility error for a flight condition is known in knots. What is done with this value to obtain the equivalent airspeed? ●
- It is subtracted from the calibrated airspeed
 - It is subtracted from the true airspeed
 - It is added to the calibrated airspeed
128. The aneroid barometer takes its name from a nerios, not fluid, because its sensing element is ●
- a curved tube with an oval cross section
 - a small capsule made of thin metal
 - a vertical tube filled with a fluid
129. What creates the negative pressure at the top of a liquid barometer tube
- The fluid flowing out of the tube due to its own weight
 - The expansion of the glass tube with temperature
 - The ambient air pressure acting on the container
130. A Bourdon tube transducer is opened for inspection. Which two parts carry out the pressure conversion? ●
- The straight tube and a position sensor
 - The curved tube and a drive coil
 - The curved tube and a position sensor
131. A Bourdon tube transducer is fitted with a synchro transmitter. Which quantity does this sensor convert into an electrical signal? ●
- The pressure inside the tube
 - The position of the tube tip
 - The temperature of the tube wall
132. How is an applied pressure distributed inside a Bourdon tube
- Equally throughout the whole tube
 - Only at the sealed tube tip
 - Only at the oval section of the tube
133. What is the amount of tube tip movement proportional to ●
- The applied pressure
 - The signal frequency
 - The tube temperature
134. Suitable bearings, clean construction of all moving parts and, in the altimeter, support with vibration are measures against ●
- backlash
 - friction
 - hysteresis
135. Of the two capsule error influences, the one that depends on whether the pressure was rising or falling is ●
- the effect of temperature
 - the elastic aftereffect
 - the hysteresis

- 136.** A diaphragm capsule delivers only a stroke of 4 to 5 mm with a small force. What must the transfer mechanism do with it? ●
- Convert it into an amplified rotation of the needle
 - Convert it into a damped rotation of the needle
 - Convert it into an amplified stroke of the capsule
- 137.** Which effect does friction have in the transfer mechanism of an aneroid instrument? ●
- It falsifies the measured result
 - It raises the capsule hysteresis
 - It damps the capsule stroke
- 138.** How is friction in the indicator mechanism kept to a minimum? ●
- With sealed bearings and grease filling of all moving parts
 - With suitable bearings and clean construction of all moving parts
 - With heavier levers and increased spring tension on all moving parts
- 139.** In the altimeter one further measure helps to reduce the effect of friction. Which measure is this? ●
- Balancing of the needle
 - Vibration of the instrument
 - Damping of the capsule
- 140.** Play in the gearing between the pinion and the tooth segment would falsify the indication. How is this play taken out? ●
- By spring tension on one side
 - By increased bearing clearance
 - By spring tension on both sides
- 141.** What is meant by the hysteresis of a membrane capsule? ●
- The stroke lags behind the pressure when a constant pressure acts for a long time
 - The curve with increasing pressure differs from the curve with increasing temperature
 - The curve with increasing pressure differs from the curve with decreasing pressure
- 142.** What causes the elastic aftereffect of a capsule? ●
- A rapidly changing pressure acting on the capsule during a climb
 - A constant pressure acting on the capsule over a longer period of time
 - A temperature change acting on the capsule mounting
- 143.** What is the result of the slow yielding of a capsule under constant pressure? ●
- A permanent offset of the needle
 - A time lag in the stroke
 - A change of the residual pressure
- 144.** Friction in the transfer mechanism, hysteresis, elastic aftereffect and temperature all work against precise operation. Which instruments do these error influences affect? ●
- Only metric scale altimeters
 - Only digital air data computers
 - All aneroid type instruments

LEVEL 3 — APPLY AND REASON

- 145.** RVSM airspace lies above FL 290 and is therefore flown on flight levels. The altimeter setting used there is ●
- the QNH of the destination airfield
 - the QFE of the departure airfield
 - the standard setting of 1013.2 hPa
- 146.** An aircraft is cleared to a level above FL 290. Which vertical separation applies there and what is the airspace called? ●
- 2000 ft separation, known as RVSM airspace
 - 1000 ft separation, known as MNPS airspace
 - 1000 ft separation, known as RVSM airspace
- 147.** An aircraft has one altimeter, an autopilot that can hold the altitude, a height deviation warning and a transponder with altitude transmission. The RVSM equipment requirement it does not meet is ●
- a transponder with altitude transmission
 - two independent altimeters
 - an autopilot for altitude hold
- 148.** An autopilot can no longer hold the selected altitude. Which requirement for flight in RVSM airspace is no longer met? ●
- A transponder with altitude transmission
 - An autopilot that can maintain the altitude
 - A device for indicating when leaving the predetermined height
- 149.** A dent is found in the skin close to a static port of an RVSM approved aircraft. Which consequence must be expected?
- A considerable influence on the altitude indication
 - A considerable influence on the mach number only
 - A considerable influence on the radio altimeter indication
- 150.** An altimeter fails in cruise and the aircraft loses its RVSM capability. Which action is required? ●
- A quick exit of RVSM airspace
 - A change of the altimeter setting to QNH
 - A descent to the transition level
- 151.** A standby airspeed indicator with bellows and the airspeed readout of the glass cockpit are read at the same moment and differ. Which airspeeds are the two showing? ●
- Calibrated airspeed on the bellows instrument and indicated airspeed on the glass cockpit
 - Indicated airspeed on the bellows instrument and calibrated airspeed on the glass cockpit
 - Indicated airspeed on the bellows instrument and true airspeed on the glass cockpit
- 152.** The indicated airspeed follows the dynamic pressure directly, and the speed formula behind it holds only for an incompressible medium. At high speed the indication is affected because air ●
- cools and generates a static pressure increase
 - compresses and generates a total pressure increase
 - expands and generates a total pressure decrease

- 153.** Navigation needs the real speed of the aircraft measured against the surrounding air, but the cockpit shows only the airspeed indication. Which speed is needed and which value must be applied to obtain it? ●
- The equivalent airspeed, obtained by applying temperature as a corrective value
 - The true airspeed, obtained by applying static pressure as a corrective value
 - The true airspeed, obtained by applying temperature as a corrective value
- 154.** Why is the equivalent airspeed lower than the calibrated airspeed at high speed and altitude? ●
- Because reduced density lowers the total pressure, so the indication is too high
 - Because compression of the air lowers the static pressure, so the indication is too low
 - Because compression of the air raises the total pressure, so the indication is too high
- 155.** Two aircraft fly at the same calibrated airspeed, one at 20,000 ft and one at 40,000 ft. Which one has the larger compressibility error?
- The aircraft at 40,000 ft
 - Both have the same error
 - The aircraft at 20,000 ft
- 156.** Static pressure acting on the capsule of an aneroid altimeter produces a stroke of only a few millimetres with a small force. The part that makes this usable for the indication is
- the position sensor, which converts it into an electrical signal
 - the transfer mechanism, which amplifies it into rotation of the needle
 - the pressure scale, which sets the reference level
- 157.** The synchro transmitter of a Bourdon tube transducer is found unserviceable while the tube itself is sound. Which part of the conversion is lost? ●
- The supply of static pressure to the curved tube
 - The deformation of the curved tube by the applied pressure
 - The conversion of the tube tip position into an electrical signal
- 158.** An altimeter needle must turn 51 times over its full range. Which demand does this place on the instrument?
- The highest demands on case sealing and static line length
 - The highest demands on material and fine mechanical precision
 - The highest demands on electrical power and signal bandwidth
- 159.** The spring that holds one side of the gearing under tension is found broken in an aneroid instrument. Which error influence is no longer compensated? ●
- Hysteresis of the membrane capsule
 - Backlash between the pinions and the tooth segments
 - Friction in the bearings of the moving parts
- 160.** Which measures are used to characterize the properties of a capsule for its intended use ●
- Special alloys, natural ageing and particular residual pressures
 - Special coatings, artificial ageing and particular bearing materials
 - Special alloys, artificial ageing and particular waveforms
- 161.** Which parts make temperature compensation demanding over the range -70 to +50 degrees C ●
- The pinion bearings and the quick release connectors of the instrument
 - The capsule mounting and the variable steering levers in the indicator mechanism
 - The capsule mounting and the flexible tubes in the static system

- 162.** An altimeter is held at one steady pressure on the test bench for a long time and the indication slowly drifts. Which error influence explains this? ●
- The elastic aftereffect of the capsule
 - The hysteresis of the capsule
 - The backlash of the pinion gearing
- 163.** An altimeter reads slightly differently at the same test pressure depending on whether the pressure was raised or lowered to reach it. Which error influence is shown? ●
- Elastic aftereffect of the diaphragm capsule
 - Friction in the transfer mechanism
 - Hysteresis of the diaphragm capsule
- 164.** Through which parts must the smallest pressure change act to deflect the needle exactly
- From the capsule via bellows, resonator and coil
 - From the diaphragm via strain gauges, bridge and amplifier
 - From the capsule via levers, shaft and cogwheel

Answers — Pressure measuring devices

- 90. a** Reduced vertical separation minimum halves the vertical spacing used up high. (L1, PDF p. 377)
- 91. c** RVSM airspace starts above FL 290 and uses 1000 ft spacing. (L1, PDF p. 377)
- 92. c** The text defines IAS as indicated airspeed. (L1, PDF p. 381)
- 93. b** Glass cockpits with an ADC display CAS. (L1, PDF p. 381)
- 94. b** The manual states TAS is required for navigation. (L1, PDF p. 383)
- 95. c** The aneroid element is a thin metal capsule. (L1, PDF p. 385)
- 96. b** The Bourdon tube arrangement turns pressure into a proportional electrical output. (L1, PDF p. 387)
- 97. a** The tube has an oval cross section in the unpressurised state. (L1, PDF p. 387)
- 98. a** The capsule stroke is only 4 to 5 mm. (L1, PDF p. 393)
- 99. c** More usable levels mean more aircraft fit into the same volume of airspace. (L2, PDF p. 377)
- 100. b** Altimeters become less precise with height, so a bigger buffer was needed. (L2, PDF p. 377)
- 101. a** The list of RVSM-sensitive components closes with the structure and surroundings of the pitot-static sensors. (L2, PDF p. 377)
- 102. a** The device warns the crew that the selected altitude is being left. (L2, PDF p. 377)
- 103. a** RVSM depends on the autopilot holding the selected altitude accurately. (L2, PDF p. 377)
- 104. a** The transponder must report altitude taken from the altitude measurement unit. (L2, PDF p. 377)
- 105. a** More available levels give better routings, saving time and fuel. (L2, PDF p. 377)
- 106. c** Improved precision of test equipment is the listed subject that deals with the equipment used to test the aircraft. (L2, PDF p. 379)
- 107. b** Approval is per aircraft and is applied for by the operator. (L2, PDF p. 379)
- 108. c** The manual says each aircraft must be modified per the required service bulletins and is subject to the altitude monitoring programme. (L2, PDF p. 379)
- 109. c** Those three are the surface waviness causes listed. (L2, PDF p. 379)
- 110. b** The training list names the part numbers of the air data computer. (L2, PDF p. 379)
- 111. c** Airworthiness approval comes first and opens the way to the test flight. (L2, PDF p. 379)
- 112. b** Operational approval is the last step and permits use of RVSM airspace. (L2, PDF p. 379)
- 113. b** IAS follows dynamic pressure, which is total pressure minus static pressure. (L2, PDF p. 381)
- 114. b** The bellows instrument is only true at mean sea level in standard conditions. (L2, PDF p. 381)
- 115. c** CAS is IAS corrected for instrument error and static source error. (L2, PDF p. 381)

- 116. a** A bellows instrument with no computer correction shows IAS. (L2, PDF p. 381)
- 117. a** The true airspeed is the one wanted for navigation; the equivalent airspeed is not indicated and serves design data and further calculations. (L2, PDF p. 383)
- 118. b** The formula assumes an incompressible medium. (L2, PDF p. 383)
- 119. b** Compression of the air at high speed raises the total pressure, which the instrument reads as extra speed. (L2, PDF p. 383)
- 120. b** The error grows with both airspeed and altitude. (L2, PDF p. 383)
- 121. b** EAS is what is left when the compressibility error is taken out of CAS. (L2, PDF p. 383)
- 122. c** EAS is a design and calculation value, not a cockpit indication. (L2, PDF p. 383)
- 123. b** TAS is the aircraft speed relative to the surrounding air mass. (L2, PDF p. 383)
- 124. a** The summary chain runs IAS to CAS to EAS to TAS. (L2, PDF p. 384)
- 125. b** The last step of the chain is the density calculation. (L2, PDF p. 384)
- 126. b** CAS becomes EAS when the compressibility error is corrected. (L2, PDF p. 384)
- 127. a** The compressibility error is taken off the calibrated airspeed of 293 knots to give the equivalent airspeed of 278 knots. (L2, PDF p. 384)
- 128. b** An aneroid barometer senses pressure with a thin metal capsule and so needs no liquid at all. (L2, PDF p. 385)
- 129. a** The weight of the fluid draws it down and leaves a negative pressure above it. (L2, PDF p. 385)
- 130. c** Only the curved oval tube and a position sensor at its tip are required. (L2, PDF p. 387)
- 131. b** The sensor senses tip position, not pressure directly. (L2, PDF p. 387)
- 132. a** The pressure acts equally over the whole tube. (L2, PDF p. 389)
- 133. a** Tip travel is proportional to the applied pressure. (L2, PDF p. 389)
- 134. b** Those three measures are the ones named for keeping friction to a minimum. (L2, PDF p. 393)
- 135. c** Hysteresis is the difference between the rising and the falling pressure curve, while the elastic aftereffect comes from pressure held constant over a long time. (L2, PDF p. 393)
- 136. a** The mechanism amplifies the tiny stroke into needle rotation. (L2, PDF p. 393)
- 137. a** Friction in the mechanism corrupts the reading. (L2, PDF p. 393)
- 138. b** Good bearings and clean mechanical construction minimise friction. (L2, PDF p. 393)
- 139. b** Vibration keeps the altimeter mechanism from sticking, helping against friction. (L2, PDF p. 393)
- 140. a** A one-sided spring tension takes up the gear backlash. (L2, PDF p. 393)
- 141. c** Hysteresis is the difference between the rising and falling pressure characteristics. (L2, PDF p. 393)
- 142. b** Long-held constant pressure makes the capsule yield slowly. (L2, PDF p. 393)
- 143. b** The stroke lags in time, which is the elastic aftereffect. (L2, PDF p. 393)
- 144. c** The manual states these errors act on all aneroid instruments. (L2, PDF p. 393)
- 145. c** RVSM airspace is flight level airspace, and flight levels are flown with the standard pressure of 1013.2 hPa set. (L3, PDF p. 377)
- 146. c** Above FL 290 the vertical separation is 1000 ft and that airspace is called RVSM airspace. (L3, PDF p. 377)
- 147. b** Three of the four required items are fitted, but RVSM needs two independent altimeters and the aircraft has only one. (L3, PDF p. 377)
- 148. b** One of the listed items of RVSM equipment is an autopilot able to maintain the altitude, and that is the function lost. (L3, PDF p. 377)
- 149. a** Skin damage near the static source disturbs the static pressure and so the altitude reading. (L3, PDF p. 377)
- 150. a** Losing RVSM capability means the aircraft must leave RVSM airspace quickly. (L3, PDF p. 379)
- 151. b** A bellows instrument shows the indicated airspeed, while the glass cockpit indication is the calibrated airspeed. (L3, PDF p. 381)
- 152. b** IAS depends directly on the dynamic pressure, and at higher speeds the air compresses and adds to the total pressure, which the formula does not allow for. (L3, PDF p. 383)

- 153. c** Navigation needs the true airspeed, and temperature is the corrective value applied so that the true airspeed can be indicated. (L3, PDF p. 383)
- 154. c** The extra total pressure from compression makes the reading too high, so the correction reduces the speed. (L3, PDF p. 383)
- 155. a** The error grows with altitude, so the higher aircraft has the larger error. (L3, PDF p. 384)
- 156. b** Static pressure moves the capsule only a few millimetres, so the transfer mechanism has to turn that small stroke into an amplified needle rotation. (L3, PDF p. 386)
- 157. c** The synchro transmitter is the position sensor, so with it unserviceable the tube tip still moves but its position is no longer turned into an electrical signal. (L3, PDF p. 387)
- 158. b** Such fine resolution needs the best material and fine mechanics. (L3, PDF p. 391)
- 159. b** The one-sided spring tension is what takes out the backlash in the gearing, so a broken spring leaves that backlash uncompensated. (L3, PDF p. 393)
- 160. c** The three listed measures are special alloys, artificial ageing and particular waveforms. (L3, PDF p. 393)
- 161. b** Compensation is hard because of the capsule mounting and the variable steering levers. (L3, PDF p. 393)
- 162. a** Slow drift under a constant held pressure is the elastic aftereffect, not hysteresis. (L3, PDF p. 393)
- 163. c** A difference between the rising and falling approach to the same pressure is hysteresis. (L3, PDF p. 393)
- 164. c** The mechanical chain runs capsule, levers, shaft, cogwheel, needle. (L3, PDF p. 393)

Pressure transducers

PDF pages 395–402 • 25 questions

LEVEL 1 — RECALL

- 165.** What is the function of a pressure transducer ●
- a. To convert measured pressures into electrical signals
 - b. To convert electrical signals into measured pressures
 - c. To convert measured pressures into mechanical needle movement
- 166.** In which equipment are pressure transducers used ●
- a. In inertial reference units
 - b. In air data computers
 - c. In flight control computers
- 167.** Several electronic pressure sensor principles exist. Which one is described as the most commonly used? ●
- a. The vibrating transducer
 - b. The piezoresistive transducer
 - c. The force compensated cell
- 168.** From which material is the micromechanical pressure transducer chamber made ●
- a. Silicon material
 - b. Pyrex material
 - c. Metal material
- 169.** What is the size of the micromechanical pressure transducer structure ●
- a. Only a few millimetres
 - b. Only a few micrometres
 - c. Only a few centimetres

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 170.** Pressure transducers are fitted in air data computers because the computer needs the measured pressures ●
- a. as an output frequency of the indicator
 - b. as amplified needle rotations
 - c. as electrical signals
- 171.** Pressure transducers have converted the measured pressures into electrical signals. What happens with these signals next?
- a. They are sent to the computer and transformed into the required output signals
 - b. They are sent to the indicator and transformed into the required needle deflection
 - c. They are sent to the probe heater and transformed into the required current
- 172.** The output frequency generated by a vibrating transducer depends on ●
- a. the input pressure
 - b. the reference chamber vacuum
 - c. the supply voltage

- 173.** Which output does a vibrating transducer generate ●
- An output frequency which depends on the reference pressure
 - An output voltage which depends on the input pressure
 - An output frequency which depends on the input pressure
- 174.** Which function does the outer case of a vibrating transducer have
- It forms the outer wall of the measured pressure chamber
 - It forms the outer wall of the reference pressure chamber
 - It forms the inner wall of the reference pressure chamber
- 175.** The type of barometer chamber the micromechanical pressure transducer is built like is ●
- the mercury type
 - the aneroid type
 - the liquid-based type
- 176.** What happens when pressure acts on the silicon lid of a piezoresistive transducer
- The lid is dented and the capacitance in its surface changes
 - The lid vibrates and the frequency of its surface changes
 - The lid is dented and the resistance in its surface changes
- 177.** What is the output voltage of a piezoresistive transducer proportional to ●
- The input pressure
 - The reference pressure
 - The supply frequency

LEVEL 3 — APPLY AND REASON

- 178.** A force compensated pressure cell is fitted instead of a plain pressure cell. Which two advantages are gained?
- It is insensitive to pressure changes and needs no separate pressure cells
 - It is insensitive to pressure changes and has small instrument errors
 - It is sensitive to pressure changes and has small instrument errors
- 179.** Which pressure sensing type is listed as needing no different pressure cells for static and pitot?
- The plain pressure cell type
 - The vibration pressure transducer
 - The force compensated pressure cell
- 180.** An electronic pressure sensor delivers an output frequency that depends on the input pressure. This sensor is ●
- the piezoresistive transducer, the most commonly used type
 - the pressure cell transducer, the most commonly used type
 - the vibrating transducer, the most commonly used type
- 181.** One transducer gives an output frequency depending on the input pressure, another an output voltage in proportion to the input pressure. These are ●
- the vibrating transducer and the micromechanical transducer respectively
 - the vibrating transducer and the Bourdon tube transducer respectively
 - the micromechanical transducer and the vibrating transducer respectively

- 182.** A vibrating transducer is cut open. Where is the pressure that is being measured found?
- In the pressure chamber inside the vibrating cylinder
 - In the pressure chamber inside the center body
 - In the reference chamber outside the vibrating cylinder
- 183.** What is done with the signal from the pickup coil
- It is amplified and used as output, and fed back to the drive coil
 - It is amplified and used as output, and fed back to the reference chamber
 - It is digitized and used as reference, and fed back to the drive coil
- 184.** Why is the pickup coil signal fed back to the drive coil of a vibrating transducer?
- To keep the output frequency at a constant value
 - To keep the reference chamber at a constant vacuum
 - To keep the cylinder vibrating with its natural frequency
- 185.** A pressure sensor consists of a chamber made of silicon, only a few millimetres in size. It is ●
- the micromechanical transducer, built as an aneroid-type chamber
 - the vibrating transducer, built as an aneroid-type chamber
 - the micromechanical transducer, built as a liquid-based chamber
- 186.** Pressure bends the diaphragm of a piezoresistive transducer and the resistance of the strain gages changes. What does the bridge then deliver, and what is it proportional to?
- An output voltage, in proportion to the input pressure
 - An output voltage, in proportion to the supply voltage
 - An output frequency, in proportion to the input pressure
- 187.** Pressure is applied to a piezoresistive transducer. How do the two bridge voltages behave?
- U1 and U2 both increase
 - U1 decreases while U2 increases
 - U1 increases while U2 decreases
- 188.** A strain gauge wire is made thicker while its length stays the same. How does its resistance change?
- It decreases with increasing area
 - It increases with increasing area
 - It is independent of the area
- 189.** A strain gauge on the bent side of the diaphragm is stretched. Why does its resistance rise?
- Because the wire becomes longer and its cross section smaller
 - Because the wire becomes shorter and its cross section larger
 - Because the material constant of the wire rises with the strain

Answers — Pressure transducers

- 165. a** Transducers turn pressure into electrical signals. (L1, PDF p. 395)
- 166. b** The manual names air data computers as an application. (L1, PDF p. 395)
- 167. a** The manual calls the vibrating transducer the most commonly used type. (L1, PDF p. 397)
- 168. a** The chamber is made of silicon. (L1, PDF p. 399)
- 169. a** The structure measures only a few millimetres. (L1, PDF p. 399)
- 170. c** A transducer turns the measured pressures into electrical signals, which is the form the air data computer works with. (L2, PDF p. 395)
- 171. a** The signals go to the computer, which makes the required outputs. (L2, PDF p. 395)
- 172. a** The frequency the vibrating transducer puts out is set by the pressure fed into it. (L2, PDF p. 397)

- 173. c** Its output is a frequency that varies with the measured pressure. (L2, PDF p. 397)
- 174. b** The outer case is the outer wall of the reference chamber. (L2, PDF p. 397)
- 175. b** Although it is made of silicon, the micromechanical transducer is described as an aneroid-type chamber. (L2, PDF p. 399)
- 176. c** Denting the lid deforms the surface resistors and changes their resistance. (L2, PDF p. 399)
- 177. a** The output voltage rises in proportion to the applied pressure. (L2, PDF p. 399)
- 178. b** The force compensated cell is insensitive to pressure changes and keeps instrument errors small. (L3, PDF p. 396)
- 179. b** The table gives 'no different pressure cells for static and pitot' for the vibration type; both pressure cell types are listed as needing different cells for static and pitot. (L3, PDF p. 396)
- 180. c** The vibrating transducer is named as the most commonly used type and it is the one whose output frequency depends on the input pressure. (L3, PDF p. 397)
- 181. a** The vibrating transducer produces a frequency; the micromechanical transducer produces a voltage in proportion to the pressure. (L3, PDF p. 397)
- 182. a** The measured pressure sits in the chamber inside the cylinder. (L3, PDF p. 397)
- 183. a** The pickup signal serves both as the amplified output and as the feedback to the drive coil. (L3, PDF p. 397)
- 184. c** The feedback loop sustains vibration at the natural frequency. (L3, PDF p. 397)
- 185. a** The micromechanical transducer is the aneroid-type chamber made of silicon whose structure is only a few millimetres in size. (L3, PDF p. 399)
- 186. a** The four strain gages sit in a wheatstone bridge, and the change in their resistance gives an output voltage in proportion to the input pressure. (L3, PDF p. 401)
- 187. c** The two halves of the bridge move in opposite directions when the plate bends. (L3, PDF p. 401)
- 188. a** A larger cross section lowers the resistance. (L3, PDF p. 401)
- 189. a** Figure 35 shows the stretched side getting longer and thinner, and $R = \rho \times l / A$, so both changes raise the resistance. (L3, PDF p. 402)

Pitot static systems

PDF pages 403–415 • 71 questions

LEVEL 1 — RECALL

- 190.** Which pressure is used by an altimeter ●
- Static pressure only
 - Pitot and static pressure
 - Pitot pressure only
- 191.** An airspeed indicator in a simple system is connected to two pressure lines. Which pressures does it use? ●
- Pitot pressure and static pressure
 - Static pressure and differential pressure
 - Pitot pressure and total pressure
- 192.** What is static pressure ●
- The actual atmospheric pressure surrounding the aircraft
 - The build-up of air pressure in a tube facing the airstream
 - The difference between total pressure and ram pressure
- 193.** A pneumatic device is calibrated to the international standard atmosphere. Which value of air density at mean sea level applies? ●
- 1,150 kg/m³
 - 1,225 kg/m³
 - 1,013 kg/m³
- 194.** What does the abbreviation SSE stand for ●
- Standard source error
 - Static source error
 - Static system error
- 195.** What is pitot pressure ●
- The build-up of air pressure in a tube with its opening facing the airstream
 - The actual atmospheric pressure in the air surrounding the aircraft
 - The build-up of air pressure in a tube with its opening across the airstream
- 196.** Which two pressures make up the total pressure P_{tot} ●
- Ram pressure and differential pressure
 - Ram pressure and static pressure
 - Static pressure and differential pressure
- 197.** Where is the pitot tube usually located on the aircraft ●
- Near the nose, on the nose itself or below the cockpit
 - Near the tail, on the fin itself or below the rudder
 - Near the nose, on the wing leading edge or below the flaps
- 198.** What does the pitot tube consist of and which pressure is present in it during the entire flight ●
- A flapper plate chamber in which total pressure P_{tot} is present
 - A cylindrical pipe in which total pressure P_{tot} is present
 - A cylindrical pipe in which static pressure P_s is present

199. Why is the Prandtl tube electrically heated ●
- To prevent condensation
 - To prevent deformation
 - To prevent icing
200. A Prandtl tube supplies both pressures from one probe. Where is it located on the aircraft? ●
- At the front of the aircraft
 - Under the wing of the aircraft
 - At the rear of the aircraft

LEVEL 2 — UNDERSTAND AND EXPLAIN

201. What the altimeter and the vertical speed indicator have in common is that ●
- Both use pitot pressure
 - Both use pitot pressure and static pressure
 - Both use static pressure
202. Why are the left and right hand static pressure measuring locations of one static system connected to each other ●
- To increase the pressure delivered to the connected instruments
 - To drain any water collected inside the static pressure lines
 - To compensate for any unbalance in pressure measurement
203. Compared with the simple system of a small aircraft, a modern commercial aircraft has
- One pitot system feeding two static systems
 - Two or three separate pitot-static pressure systems
 - One pitot system and one static system as well
204. What does the alternate static source selector valve allow the Captain to do ●
- Use the copilot's static pressure system
 - Isolate the copilot's static pressure system
 - Use the copilot's pitot pressure system
205. Static ports are made flush with the skin of the aircraft so that
- Water entering the ports is drained away again
 - The ports can be heated electrically from inside the skin
 - The surface stays smooth and no pressure distorting turbulences occur
206. On which two quantities does air pressure p depend ●
- Air density and temperature
 - Temperature and angle of attack
 - Air density and mach number
207. What do aircraft manufacturers ensure when they choose the position of the static ports ●
- That they lie in areas with low temperature and air density variation
 - That they lie in areas with high turbulence and ram air distortion
 - That they lie in areas with low turbulence and ram air distortion
208. Why do most helicopters have special cover plates at the static ports ●
- To compensate the effects caused by the strong vibration of the rotor
 - To compensate the effects caused by the strong downwash of the rotor
 - To compensate the effects caused by the high angle of attack of the rotor

209. Among the factors the static source error depends on, the one qualified as applying only where relevant is ●
- The position of the flaps
 - The mach number
 - The angle of attack
210. Why can a static source error arise on an aircraft ●
- It is not always possible to calibrate the pneumatic devices to the ISA values
 - It is not always possible to keep the static ports electrically heated in flight
 - It is not always possible to find turbulence and ram air free zones on the aircraft
211. An engineer is told that the static source error of a jet aeroplane is not a constant value. On which factors does it depend? ●
- The mach number, the angle of attack, the rudder position and the nose gear position
 - The mach number, the air density, the flap position and the outside air temperature
 - The mach number, the angle of attack, the flap position and the nose gear position
212. How large is the static source error on slow-flying propeller planes ●
- So large that it must be corrected
 - So small that it must be measured
 - So small that it is negligible
213. The crew has to obtain the correct value from the distorted values being displayed. Where are the correction tables found?
- In the Aircraft Maintenance Manual or in the flight log
 - In the Operating Manual or in the weight and balance data
 - In the Operating Manual or in the flight regulations
214. Which pressure reading is affected by a static source error ●
- The ram pressure reading
 - The static pressure reading
 - The total pressure reading
215. The static source error is listed with several dependencies. Which of them is marked as unusual? ●
- The position of the nose gear
 - The position of the flaps
 - The angle of attack
216. What can happen to the static source error of a jet aeroplane flying near the speed of sound ●
- It can fall below the tolerances required by law
 - It can become independent of the angle of attack
 - It can exceed the tolerances required by law
217. The static pressure has to be known before the ram pressure can be obtained because ●
- Static pressure and ram pressure are measured by the same probe
 - Total pressure is made up of ram pressure plus static pressure
 - Ram pressure is the atmospheric pressure surrounding the aircraft
218. The difference between total pressure and ram pressure is that ●
- Total pressure also contains the dynamic pressure
 - Total pressure also contains the static pressure
 - Total pressure is measured at the static ports

- 219.** Which pressure alone is required to measure the speed of the aircraft in relation to the ambient air ●
- The ram pressure q
 - The total pressure P_{tot}
 - The static pressure p_s
- 220.** How is ram pressure q obtained in the instrument ●
- By measuring the static pressure at two separate static ports
 - By measuring the sum of the total pressure and the static pressure
 - By measuring the differential pressure between total and static pressure
- 221.** Ram pressure cannot be sensed on its own. Which pressure is already known from the static ports and is used to obtain it? ●
- The dynamic pressure q
 - The total pressure P_{tot}
 - The static pressure p_s
- 222.** Comparing the electrical heating of the pitot tube with that of the static ports
- The pitot tube is heated, while only some aircraft types have heated static ports
 - Both are heated on every aircraft type
 - Only the static ports are heated, the pitot tube being self draining
- 223.** A pitot tube is aligned so that it faces the predetermined airstream because ●
- The airstream stops the tube from freezing up
 - Pitot pressure builds up in a tube whose opening faces the airstream
 - The airstream keeps the drain hole clear of water
- 224.** A pitot tube is being refitted to an aeroplane after removal. How must it be aligned? ●
- So that it faces the predetermined rotor downwash
 - So that it faces the predetermined airstream
 - So that it lies across the predetermined airstream
- 225.** What do the small drain hole and the baffle of a pitot tube ensure ●
- That the tube inlet is kept free of any deformation
 - That ram air entering the tube is slowed down
 - That water entering the tube is drained away
- 226.** A student is worried that the drain hole in a pitot tube lets pressure escape. Which statement from the manual answers this? ●
- The drain hole and the baffle do not affect the total pressure
 - The drain hole and the baffle slightly reduce the total pressure
 - The drain hole and the baffle slightly reduce the static pressure
- 227.** The Prandtl tube is called a pitot-static-tube because ●
- It combines a pitot tube with a sensor to measure static pressure
 - It is heated electrically like the static ports
 - It is fitted at the front of the aircraft like a pitot tube
- 228.** Comparing where the Prandtl tube and the pitot tube are fitted ●
- The Prandtl tube is located on the sides of the fuselage and the pitot tube at the front
 - The Prandtl tube is located below the rotor and the pitot tube at the front
 - The Prandtl tube is located in the same way as the pitot tube at the front of the aircraft

- 229.** On a Prandtl tube the cylindrical inlet and the holes drilled in the sides supply, in that order ●
- The connected static lines and the total pressure
 - The total pressure and the connected static lines
 - The total pressure and the drain line
- 230.** What does the Prandtl tube combine in one unit ●
- A pitot tube with a sensor to measure total air temperature
 - A pitot tube with a sensor to measure static pressure
 - A static port with a sensor to measure the angle of attack
- 231.** A Prandtl tube has a cylindrical inlet at the front and a row of holes drilled in its sides. What do these side holes supply? ●
- Information for one or two connected pitot lines
 - Information for one or two connected static lines
 - Drainage for one or two connected water traps
- 232.** What do the flapper plate and drain hole of a Prandtl tube prevent ●
- Icing from forming at the tube inlet
 - Water from entering the pitot line
 - Roughness from forming on the tube surface
- 233.** A Prandtl tube is found with a rough surface after a repair. Which consequence does the manual warn about?
- It will lead to total pressure errors
 - It will lead to static pressure errors
 - It will lead to heating element failures
- 234.** How does a Prandtl tube obtain the ram pressure and static pressure needed for speed indication ●
- From two separate measuring points
 - From two connected static ports
 - From one single ram air tube
- 235.** Which two separate measuring points does the Prandtl tube replace ●
- The pitot tube and the static port
 - The static port and the difference bellows
 - The pitot tube and the air data computer
- 236.** A Prandtl tube delivers total pressure and static pressure. Where is the required differential pressure obtained? ●
- In the static port or in the drain hole
 - In the tube itself or in the piping
 - In the instrument or in a computer

LEVEL 3 — APPLY AND REASON

- 237.** The pitot probe of a simple system becomes completely blocked while the static system stays clear. Which of the three supplied instruments is affected? ●
- The airspeed indicator
 - The vertical speed indicator
 - The altimeter

- 238.** Which instrument of the simple system needs a pressure connection in addition to the static system ●
- The vertical speed indicator
 - The airspeed indicator
 - The altimeter
- 239.** During a repair the connection between the left and right hand measuring locations of one static pressure system is left open. What is lost by this? ●
- The supply of pitot pressure to the instruments
 - The heating of the static measuring locations
 - The compensation for any unbalance in pressure measurement
- 240.** A technician traces one static pressure system of a small aeroplane and finds a measuring location on the left and on the right hand side of the fuselage. Which statement describes this arrangement correctly? ●
- Both locations belong to the same static system and are connected together
 - Each location belongs to a separate static system and they stay isolated
 - Both locations belong to the same pitot system and are connected together
- 241.** The Captain moves the handle of the alternate static source selector valve to its other position. Which static pressure then reaches his instruments? ●
- The pitot pressure of his own system
 - His own static pressure system
 - The copilot's static pressure system
- 242.** An aeroplane has a fault in one pilot's static system and is not one of the few types that can be switched to the auxiliary system. Which possibility remains?
- Use of the other pilot's static system via the selector valve
 - Use of the pitot system via the selector valve
 - Use of the auxiliary system via the selector valve
- 243.** A helicopter cannot keep its static ports in air that is free of rotor downwash. Which fitting deals with this? ●
- Special cover plates at the static ports
 - Electrically heated static ports
 - A drain hole at each static port
- 244.** After a skin repair the zone around a static port is left with a rough surface. What must be expected at that port?
- A blockage of the pressure pipe
 - A loss of the port heating
 - Pressure distorting turbulences
- 245.** Why are static ports placed in areas with low turbulence and ram air distortion ●
- Because turbulence and ram air distortion falsify the static pressure reading
 - Because turbulence and ram air distortion overheat the static port zones
 - Because turbulence and ram air distortion block the static pressure hoses
- 246.** An aeroplane extends its flaps on the approach. Which reading can change as a result of this alone? ●
- The static pressure reading
 - The angle of attack reading
 - The total air temperature reading

247. Of the factors the static source error depends on, the two that change with the configuration of the aircraft are ●
- The mach number and the position of the flaps
 - The mach number and the angle of attack
 - The position of the flaps and the position of the nose gear
248. An aeroplane changes its angle of attack in cruise with everything else unchanged. Which measurement may be affected by this? ●
- The total pressure measurement
 - The pitot pressure measurement
 - The static pressure measurement
249. A slow-flying propeller plane and a jet aeroplane near the speed of sound differ in one of the listed static source error factors above all. Which factor is that? ●
- The position of the flaps
 - The mach number
 - The position of the nose gear
250. A jet aeroplane flies near to the speed of sound and its static source error exceeds the tolerances required by law. Which correction becomes necessary? ●
- A correction using an air data computer
 - A correction using a differential pressure bellows
 - A correction using an alternate static source valve
251. A static source error falsifies the static pressure supplied to a simple pitot-static system. Which instruments are affected? ●
- Only the airspeed indicator of the system
 - All three instruments of the simple system
 - Only the vertical speed indicator of the system
252. A slow-flying propeller aeroplane and a jet aeroplane near the speed of sound both have a static source error. Which comparison is correct? ●
- Negligible on the propeller aeroplane and possibly beyond the legal tolerance on the jet
 - Negligible on the jet aeroplane and possibly beyond the legal tolerance on the propeller plane
 - Beyond the legal tolerance on both aeroplanes and corrected by the air data computer
253. In the relation $P_{tot} = q + p_s$, the quantity p_s is delivered to the instrument from ●
- The pitot probe
 - The static ports
 - The drain hole of the pitot tube
254. An airspeed indicator is fed from the pitot probe as well as the static system because ●
- The static pressure alone gives the speed in the surrounding air
 - The pitot probe cannot deliver a pressure without a static reference
 - The ram pressure it needs is the total pressure less the static pressure
255. A pitot tube delivers the pressure present in its cylindrical pipe throughout the flight. What must be supplied in addition before a speed can be obtained? ●
- The static pressure from the static ports
 - The total air temperature from a temperature probe
 - A second total pressure from another pitot tube

- 256.** An aircraft type is to take both the ram pressure and the static pressure needed for speed indication from one single ram air tube. Which probe is used? ●
- A pitot tube
 - A Prandtl tube
 - A multi function probe
- 257.** A Prandtl tube carries a feature that a plain pitot tube does not, namely ●
- Electrical heating of the tube
 - A cylindrical inlet for the total pressure
 - Holes drilled in the sides supplying connected static lines
- 258.** Both a pitot tube and a Prandtl tube keep water out of the pitot line. Which item does this in the Prandtl tube together with the drain hole? ●
- A baffle
 - A cover plate
 - A flapper plate
- 259.** On an aircraft type fitted with a Prandtl tube the static pressure needed for speed indication comes from ●
- A pair of static ports on the sides of the fuselage
 - The same single ram air tube that supplies the ram pressure
 - The cylindrical inlet at the front of the tube
- 260.** A Prandtl tube delivers total pressure and static pressure to a computer. Which quantity does the computer obtain from them, and what is it needed for? ●
- The total air temperature, needed for the speed indication
 - The static source error, needed for the altitude indication
 - The ram pressure, needed for the speed indication

Answers — Pitot static systems

- 190. a** The manual states the altimeter works on static pressure alone. (L1, PDF p. 403)
- 191. a** The airspeed indicator is the only one of the three that needs both pitot and static pressure. (L1, PDF p. 403)
- 192. a** Static pressure is simply the ambient atmospheric pressure around the aircraft. (L1, PDF p. 407)
- 193. b** The ISA sea level density given in the manual is 1,225 kg/m³. (L1, PDF p. 407)
- 194. b** SSE is the static source error. (L1, PDF p. 409)
- 195. a** Pitot pressure builds up in a tube whose opening points into the airstream. (L1, PDF p. 411)
- 196. b** The manual gives total pressure as ram pressure plus static pressure. (L1, PDF p. 411)
- 197. a** These are the three locations the manual names. (L1, PDF p. 413)
- 198. b** It is a cylindrical pipe carrying total pressure throughout the flight. (L1, PDF p. 413)
- 199. c** Electrical heating keeps the probe free of ice. (L1, PDF p. 415)
- 200. a** It sits at the front, like the pitot tube. (L1, PDF p. 415)
- 201. c** Static pressure is named as the pressure used by each of these two instruments. (L2, PDF p. 403)
- 202. c** Cross-connecting the two sides evens out an unbalanced pressure reading between them. (L2, PDF p. 403)
- 203. b** The simple system has one of each, whereas the modern commercial aircraft has two or three separate systems. (L2, PDF p. 405)
- 204. a** The valve switches the Captain over to the copilot's static source. (L2, PDF p. 405)
- 205. c** A flush port keeps the zone smooth, which is what the warning requires so that no pressure distorting turbulences arise. (L2, PDF p. 407)
- 206. a** The manual lists air density and temperature as the two quantities. (L2, PDF p. 407)

- 207. c** Low turbulence and low ram air distortion keep the static reading true. (L2, PDF p. 407)
- 208. b** The cover plates offset the pressure effects of the rotor downwash. (L2, PDF p. 407)
- 209. c** The angle of attack is the item carrying the qualification if applicable. (L2, PDF p. 409)
- 210. c** Where no turbulence free and ram air free zone exists, the sensed static pressure is falsified. (L2, PDF p. 409)
- 211. c** These four items are the dependencies the manual lists for the SSE. (L2, PDF p. 409)
- 212. c** At low speed the error is negligible. (L2, PDF p. 409)
- 213. c** The manual names the Operating Manual and the flight regulations as the sources of the correction tables. (L2, PDF p. 409)
- 214. b** The SSE falsifies the static pressure reading. (L2, PDF p. 409)
- 215. a** Only the nose gear position carries the remark unusual. (L2, PDF p. 409)
- 216. c** Close to the speed of sound the error may go beyond the legal tolerance. (L2, PDF p. 409)
- 217. b** Since total pressure is the sum of ram and static pressure, the static part must be known so that it can be taken out. (L2, PDF p. 411)
- 218. b** Total pressure is ram pressure together with static pressure, so the static part is what it holds in addition. (L2, PDF p. 411)
- 219. a** Only the dynamic or ram pressure q is needed for speed measurement. (L2, PDF p. 411)
- 220. c** Ram pressure is the difference between total and static pressure. (L2, PDF p. 411)
- 221. c** The static ports supply p_s , which is subtracted from P_{tot} to leave q . (L2, PDF p. 411)
- 222. a** Heating is stated plainly for the pitot tube, whereas heated static ports are stated only for some aircraft types. (L2, PDF p. 413)
- 223. b** Pitot pressure only builds up when the opening of the tube faces the airstream, so the alignment is what produces the pressure. (L2, PDF p. 413)
- 224. b** The tube must point into the airstream to build up total pressure. (L2, PDF p. 413)
- 225. c** Their job is to get rid of water that has entered the tube. (L2, PDF p. 413)
- 226. a** The manual states in brackets that they do not affect the total pressure. (L2, PDF p. 413)
- 227. a** The name follows from the tube holding both a pitot tube and a static pressure sensor in one unit. (L2, PDF p. 415)
- 228. c** The two probes are fitted the same way, at the front of the aircraft. (L2, PDF p. 415)
- 229. b** Total pressure enters at the cylindrical inlet while the side holes feed the static lines. (L2, PDF p. 415)
- 230. b** It puts the pitot function and a static pressure sensor into one probe. (L2, PDF p. 415)
- 231. b** The side holes feed the static lines, while the front inlet feeds total pressure. (L2, PDF p. 415)
- 232. b** They keep water out of the pitot line. (L2, PDF p. 415)
- 233. b** Roughness disturbs the flow over the side holes and falsifies static pressure. (L2, PDF p. 415)
- 234. c** The Prandtl tube takes both pressures at one single tube. (L2, PDF p. 415)
- 235. a** The manual names the pitot tube and the static port as the two points it replaces. (L2, PDF p. 415)
- 236. c** The subtraction is done in the instrument or in a computer, not in the probe. (L2, PDF p. 415)
- 237. a** Of the three instruments only the airspeed indicator uses pitot pressure, so only it depends on the pitot probe. (L3, PDF p. 403)
- 238. b** The altimeter and vertical speed indicator use static pressure only, while the airspeed indicator also needs pitot pressure. (L3, PDF p. 403)
- 239. c** The two locations are joined only to compensate unbalance in pressure measurement, so breaking the joint removes that compensation. (L3, PDF p. 403)
- 240. a** One static system uses both side locations and they are joined to each other. (L3, PDF p. 403)
- 241. c** The valve exists so that the Captain can take his static pressure from the copilot's system instead of his own. (L3, PDF p. 405)
- 242. a** Switching to the auxiliary system is limited to a few types, so the cross-over to the other pilot's static system is what remains. (L3, PDF p. 405)

- 243. a** The cover plates are fitted for exactly this purpose, to compensate the effects of the strong rotor downwash at the ports. (L3, PDF p. 407)
- 244. c** The zone is required to be smooth precisely so that no pressure distorting turbulences occur, so a rough surface allows them. (L3, PDF p. 407)
- 245. a** Where such zones cannot be found the static pressure reading is affected, so a quiet zone keeps the reading true. (L3, PDF p. 407)
- 246. a** The static source error depends on the position of the flaps, and it is the static pressure reading that this error affects. (L3, PDF p. 409)
- 247. c** Flaps and nose gear are the two listed items that are set by the aircraft configuration, the other two being flight condition values. (L3, PDF p. 409)
- 248. c** The angle of attack is one of the factors the static source error depends on, and that error affects the static pressure reading. (L3, PDF p. 409)
- 249. b** The two cases are set apart by how near the speed of sound they fly, which is the mach number among the listed factors. (L3, PDF p. 409)
- 250. a** The manual calls for an air data computer correction in this case. (L3, PDF p. 409)
- 251. b** All three instruments of the simple system are supplied with static pressure, so all three are affected. (L3, PDF p. 409)
- 252. a** The manual calls the error negligible at low speed and possibly beyond the legal tolerance near the speed of sound. (L3, PDF p. 409)
- 253. b** The static pressure p_s used in the relation is the pressure measured at the static ports. (L3, PDF p. 411)
- 254. c** Speed needs the ram pressure, and the ram pressure is obtained by subtracting the static pressure from the total pressure delivered by the pitot probe. (L3, PDF p. 411)
- 255. a** The pitot tube gives total pressure only, and the static pressure from the static ports has to be subtracted from it to get the ram pressure. (L3, PDF p. 413)
- 256. b** The Prandtl tube is the single ram air tube that supplies both of these pressures. (L3, PDF p. 415)
- 257. c** Both probes have a cylindrical pipe and electrical heating, but the side holes feeding static lines belong to the Prandtl tube. (L3, PDF p. 415)
- 258. c** The Prandtl tube works with a flapper plate and a drain hole, the baffle being the item named for the pitot tube. (L3, PDF p. 415)
- 259. b** On these types the two pressures come from the one Prandtl tube instead of separate pitot tube and static port. (L3, PDF p. 415)
- 260. c** The differential of the two delivered pressures is the ram pressure q , which is what the speed indication requires. (L3, PDF p. 415)

Pitot static system leaks

PDF pages 417–429 · 31 questions

LEVEL 1 — RECALL

- 261.** An air data system receives three parameters from one single multi function probe. Which parameters are they? ●
- Angle of sideslip, total air temperature and total pressure
 - Angle of attack, static air temperature and static pressure
 - Angle of attack, total air temperature and total pressure
- 262.** Which part of the multi function probe measures the angle of attack ●
- A wind vane
 - A pitot probe
 - A TAT tube
- 263.** The colour used to designate pitot system connections ●
- Red
 - Blue
 - Yellow
- 264.** Two lines are parted at a quick-disconnect during maintenance. What happens at the coupling? ●
- It seals itself
 - It stays open until a cap is fitted
 - It vents the line to ambient air
- 265.** A technician must check the mouth of a static port against its limits. Where are these limits normally found? ●
- In ATA chapter 31
 - In ATA chapter 34
 - In ATA chapter 21
- 266.** A technician must assess damage to the skin next to a static port. Which equipment is used? ●
- A scale and a feeler-gage
 - A straight edge and a torque wrench
 - A scale and a depth micrometer
- 267.** The system that controls the cabin rate ●
- The pitot static system
 - The anti-ice system
 - The airconditioning system

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 268.** A multi function probe must deliver the total pressure to the air data system. Which of its parts provides this parameter? ●
- The pitot probe
 - The temperature tube
 - The wind vane

- 269.** Each multi function probe has to pass its measured data to the related computer. Through which item does it do so?
- a. Through the digital converter
 - b. Through the difference bellows
 - c. Through the air data computer
- 270.** In some aircraft types the two pneumatic back up instruments are arranged so that
- a. Only the altitude function is retained
 - b. Both functions are shared between three instruments
 - c. Both functions are integrated in one instrument
- 271.** An instrument has been changed in the pitot static system. What must be carried out? ●
- a. An approved flushing of the static system
 - b. An approved accuracy check of the static ports
 - c. An approved leak-proof test
- 272.** The purpose of the scale and the feeler-gage used at a static port ●
- a. To measure the mouth of the static port against the limits
 - b. To check that the static port zone is smooth enough to prevent turbulences
 - c. To assess damage of the aircraft skin against the limits
- 273.** The reason why the static port zones must have a smooth surface ●
- a. So that no turbulences can occur
 - b. So that no ice can form on the port
 - c. So that water can drain away from the port
- 274.** Where the limits for damage of the aircraft skin at the static ports are found ●
- a. In the ATA chapter for the aircraft structure or the SRM
 - b. In the ATA chapter for air conditioning or the wiring manual
 - c. In the ATA chapter for the instruments or the IPC
- 275.** What the results of a leakage depend on
- a. The position of the leakage and the outside air temperature
 - b. The size of the leakage and the altitude of the aircraft
 - c. The size of the leakage and whether the cabin is pressurized
- 276.** A static line leaks inside the unpressurized cabin of an aircraft in flight. What does the altimeter do? ●
- a. It slightly increases the indicated altitude
 - b. It slightly decreases the indicated altitude
 - c. It freezes at the altitude flown when the leak started
- 277.** The effect of a static system leakage inside an unpressurized cabin on the airspeed indicator ●
- a. It shows a slightly lower indicated airspeed
 - b. It shows zero indicated airspeed
 - c. It shows a slightly higher indicated airspeed
- 278.** The reason why the altimeter and the airspeed indicator decrease their indication when a static line leaks into a pressurized cabin ●
- a. The pitot and the static pressure become nearly the same
 - b. The pressure in the static lines increases
 - c. The pressure in the static lines decreases

- 279.** The effect of a static system leakage into a pressurized cabin on the pressure in the static lines ●
- The pressure increases
 - The pressure stays unchanged
 - The pressure decreases
- 280.** The indications of the altimeter and the airspeed indicator after a static leakage into a pressurized cabin ●
- Both stay unchanged
 - Both increase
 - Both decrease

LEVEL 3 — APPLY AND REASON

- 281.** An air data system is to receive the angle of attack from the same probe that gives it the total pressure. Which probe meets this? ●
- A Prandtl tube
 - A pitot tube
 - A multi function probe
- 282.** A multi function probe delivers the total pressure to the air data system, so the quantity still required for a speed indication is ●
- The static pressure
 - The angle of attack
 - The total air temperature
- 283.** An aircraft with a modern EFIS carries two pneumatic back up instruments. Which of them has to be connected to the pitot system? ●
- Both of them equally
 - The standby altimeter
 - The standby airspeed indicator
- 284.** The standby altimeter fitted as a back up in an EFIS aircraft has to be supplied with ●
- Static pressure only
 - Pitot pressure only
 - Pitot pressure and static pressure
- 285.** A device is replaced in lines with screw connections whose connectors are designated red. The system concerned and the action required afterwards ●
- The pitot system, and the self-sealing connections make a test unnecessary
 - The pitot system, and an approved leak-proof test must be carried out
 - The static system, and an approved leak-proof test must be carried out
- 286.** A line was parted only at a quick-disconnect. What applies to the leak-proof test?
- It is required, but only on the static side
 - It is required, as after any other parting
 - It is not required in this case

- 287.** A technician must check the mouth of a static port and also damage to the skin around it. Where are the two sets of limits found? ●
- The port mouth limits in ATA chapter 34 and the skin damage limits in the structure chapter or the SRM
 - The port mouth limits in the structure chapter or the SRM and the skin damage limits in ATA chapter 34
 - Both the port mouth limits and the skin damage limits in ATA chapter 34
- 288.** A static line breaks inside an unpressurized cabin during flight. How does the vertical speed indicator behave? ●
- It jumps once and then stays at zero
 - It jumps once and then shows correct data
 - It slowly drifts into a permanent climb indication
- 289.** The same static line leak is considered in an unpressurized aircraft and in a pressurized one. How does the altimeter behave in each case? ●
- It decreases the indication in the unpressurized aircraft and slightly increases it in the pressurized one
 - It slightly increases the indication in the unpressurized aircraft and decreases it in the pressurized one
 - It slightly increases the indication in both aircraft
- 290.** The behaviour of the vertical speed indicator after a static system leakage in an unpressurized cabin compared with a pressurized cabin ●
- An indication depending on cabin rate in the unpressurized cabin, and a small jump and then correct data in the pressurized cabin
 - A small jump and then correct data in the unpressurized cabin, and an indication depending on cabin rate in the pressurized cabin
 - A small jump and then correct data in both cabins
- 291.** A static line leaks into a pressurized cabin. What does the vertical speed indication now depend on? ●
- The cabin rate
 - The size of the leak
 - The aircraft rate of climb

Answers — Pitot static system leaks

- 261. c** AOA, TAT and total pressure are the three outputs of the probe. (L1, PDF p. 417)
- 262. a** The wind vane is the AOA sensing element. (L1, PDF p. 417)
- 263. a** Pitot connections carry the red designation; yellow belongs to the static side. (L1, PDF p. 423)
- 264. a** A self-sealing coupling closes the line itself the moment it is parted. (L1, PDF p. 423)
- 265. b** The manual points to ATA chapter 34 for the static port mouth limits. (L1, PDF p. 425)
- 266. a** The manual names the scale and the feeler-gage as the assessment tools. (L1, PDF p. 425)
- 267. c** The manual names the airconditioning system as the controller of cabin rate. (L1, PDF p. 429)
- 268. a** The built-in pitot probe supplies the total pressure. (L2, PDF p. 417)
- 269. a** The digital converter is the link between the probe and the related computer. (L2, PDF p. 417)
- 270. c** The two standby functions can be brought together in a single instrument on some types. (L2, PDF p. 421)
- 271. c** Changing an instrument or a device calls for an approved leak-proof test. (L2, PDF p. 423)
- 272. c** The scale and feeler-gage are the tools named for assessing skin damage against the AMM limits. (L2, PDF p. 425)
- 273. a** A smooth zone keeps the airflow undisturbed so the port senses true static pressure. (L2, PDF p. 425)
- 274. a** Skin damage is a structural item, so the structure chapter or the SRM holds those limits. (L2, PDF p. 425)
- 275. c** Size of the leak and the pressurization state of the cabin are the two factors given. (L2, PDF p. 427)

- 276. a** The lower cabin pressure reaching the static line reads as a slightly greater altitude. (L2, PDF p. 427)
- 277. c** The airspeed indicator reads slightly high, in the same sense as the altimeter. (L2, PDF p. 427)
- 278. b** Cabin pressure entering the static lines raises the pressure there, and both instruments read lower as a result. (L2, PDF p. 429)
- 279. a** Cabin pressure is above the ambient static pressure, so the static lines see a rise. (L2, PDF p. 429)
- 280. c** The raised static pressure drives both indications down. (L2, PDF p. 429)
- 281. c** Only the multi function probe is stated to give angle of attack as well as total pressure. (L3, PDF p. 417)
- 282. a** The ram pressure needed for speed is the total pressure less the static pressure, and the probe supplies only the total pressure of the two. (L3, PDF p. 417)
- 283. c** An airspeed indicator uses pitot pressure as well as static pressure, while an altimeter uses static pressure only. (L3, PDF p. 421)
- 284. a** It is a pneumatic altimeter, and an altimeter works on static pressure alone. (L3, PDF p. 421)
- 285. b** Red designates the pitot connections, and changing an instrument or a device calls for an approved leak-proof test. (L3, PDF p. 423)
- 286. c** A parting at a quick-disconnect is the one exception to the leak-proof test requirement. (L3, PDF p. 423)
- 287. a** The mouth of the port is covered by ATA chapter 34, while skin damage is covered by the structure chapter or the SRM. (L3, PDF p. 425)
- 288. b** The VSI reacts only to the sudden change and then works normally again. (L3, PDF p. 427)
- 289. b** On PDF page 432 the leak into an unpressurized cabin lowers the pressure in the lines, so the altimeter reads slightly higher; on this page the cabin pressure raises the line pressure, so it reads lower. (L3, PDF p. 429)
- 290. b** On PDF page 432 the unpressurized case gives a single small jump and then correct data; on this page the pressurized cabin makes the indication follow the cabin rate. (L3, PDF p. 429)
- 291. a** With cabin air in the static line the VSI follows the cabin rate, not the aircraft. (L3, PDF p. 429)

Pitot static blockages and icing

PDF pages 431–443 · 41 questions

LEVEL 1 — RECALL

- 292.** The instrument affected by a leakage in the pitot system ●
- The altimeter only
 - The vertical speed indicator only
 - The airspeed indicator only
- 293.** The indications of the altimeter and the vertical speed indicator during a climb with a blocked static system ●
- The altimeter holds the same altitude and the vertical speed indication rises
 - The altimeter holds the same altitude and the vertical speed indication stays at zero
 - The altimeter increases and the vertical speed indication stays at zero
- 294.** The indication of the vertical speed indicator during a descent with a blocked static system ●
- A steady rate of descent
 - A zero rate
 - A steady rate of climb
- 295.** What a completely blocked pitot tube means ●
- The drain hole is blocked and the inlet is open
 - The inlet and the drain hole are blocked
 - The inlet is blocked and the drain hole is open
- 296.** A pitot tube is completely blocked by ice. Which instruments are not affected? ●
- The altimeter and the airspeed indicator
 - The altimeter and the vertical speed indicator
 - The airspeed indicator and the vertical speed indicator
- 297.** The pitot inlet is blocked by ice but the water drain hole is still open. What does the airspeed indication do? ●
- It decreases to zero
 - It increases into the overspeed region
 - It stays unchanged at the last value
- 298.** The indications fluctuate during a check of the pitot static system. What does this show? ●
- The presence of ice at the static port
 - The presence of water in them
 - The presence of a leak in them

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 299.** A pitot line leaks inside an unpressurized cabin. What does the airspeed indication do? ●
- It decreases to zero
 - It increases into the overspeed region
 - It stays at the value flown when the leak started
- 300.** The reason why the airspeed indication decreases when both the pitot and the static system leak ●
- Pitot and static pressure tend to be the same
 - The total pressure in the pitot line is trapped at a constant value
 - The static pressure in the lines rises above the pitot pressure

- 301.** The effect on the airspeed indication of a leakage in both the pitot and the static system ●
- It is not affected
 - It decreases
 - It increases
- 302.** Both the pitot system and the static system are leaking. What do the altimeter and the vertical speed indicator show? ●
- Correct indications, because the two leaks cancel each other
 - The same results as for a leakage in the static system
 - The same results as for a leakage in the pitot system
- 303.** The indications with a blocked static system while the aircraft flies at constant altitude and constant airspeed ●
- The airspeed indication slowly decreases
 - There are no changes in the indication
 - The altitude indication slowly increases
- 304.** The static system is blocked and the aircraft climbs. What does the airspeed indicator show? ●
- An increasing airspeed indication
 - Zero airspeed
 - A decreasing airspeed indication
- 305.** The indications of the altimeter and the vertical speed indicator with a blocked static system, during a climb and during a descent ●
- The same altitude and a vertical speed of zero
 - The same altitude and a vertical speed that follows the aircraft
 - A changing altitude and a vertical speed of zero
- 306.** The static system is blocked and the aircraft descends. What does the airspeed indicator show? ●
- A decreasing airspeed indication
 - An increasing airspeed indication
 - Zero airspeed
- 307.** The reason why the airspeed indicator reacts like an altimeter when the pitot tube is totally blocked ●
- Total and static pressure become nearly equal, so the indication falls away
 - The static pressure is trapped, so only the changing total pressure moves the indication
 - The total pressure is trapped, so only the changing static pressure moves the indication
- 308.** The indication of the airspeed indicator during a climb with a totally blocked pitot tube ●
- It increases and can reach the overspeed region
 - It stays at the speed flown when the blockage occurred
 - It decreases and can reach zero
- 309.** The behaviour of the airspeed indicator when the pitot tube is totally blocked ●
- It reacts like a vertical speed indicator
 - It reacts like an altimeter
 - It reacts like a turn coordinator
- 310.** The indication of the airspeed indicator during a descent with a totally blocked pitot tube ●
- It decreases to a lower indicated airspeed
 - It remains at the value held during cruise
 - It increases to a higher indicated airspeed

- 311.** The pitot pressure when the pitot tube inlet is blocked but the water drain hole is still open ●
- It stays at the value it had before the blockage
 - It increases to the value of the total pressure
 - It decreases to the value of the static pressure
- 312.** The reason why the altimeter and the vertical speed indicator are not affected by a blocked pitot inlet ●
- They get both pitot and static pressure
 - They only get pitot pressure
 - They only get static pressure
- 313.** The reason why water can be present in the pressure pipes ●
- They are not drained on a regular basis
 - They are capped for protection during testing
 - Their test fittings are used to simulate altitude and speed
- 314.** The static ports of a cold aeroplane are capped and the aircraft is moved into a warm hangar. What happens in the static system?
- The pressure increases
 - The pressure decreases
 - The pressure stays the same
- 315.** The ram pressure when both the pitot inlet and the drain are iced over ●
- It depends on the altitude
 - It is always zero
 - It is constant at every altitude

LEVEL 3 — APPLY AND REASON

- 316.** A pitot line is leaking inside the cabin. How does the airspeed indication behave in an unpressurized aircraft compared with a pressurized one? ●
- It is unpredictable in the unpressurized aircraft and decreases to zero in the pressurized one
 - It decreases to zero in both aircraft
 - It decreases to zero in the unpressurized aircraft and is unpredictable in the pressurized one
- 317.** The reason why the airspeed indication falls to zero with a pitot leakage in an unpressurized cabin ●
- The static pressure becomes much greater than the pitot pressure
 - Static and pitot pressure become nearly the same
 - The pitot pressure rises with the cabin pressure
- 318.** A pitot line leaks inside a pressurized cabin. What can be said about the airspeed indication? ●
- It always falls to zero, because pitot and static pressure become equal
 - It is unpredictable, because it depends on altitude and cabin pressure
 - It always rises, because the cabin pressure is higher than the static pressure
- 319.** The instruments disturbed by a leakage in the pitot system alone compared with a leakage in both the pitot and the static system ●
- The airspeed indicator alone, compared with the airspeed indicator, the altimeter and the vertical speed indication
 - The airspeed indicator and the vertical speed indication, compared with the airspeed indicator alone
 - The airspeed indicator and the altimeter, compared with the airspeed indicator and the vertical speed indication

- 320.** Both the pitot system and the static system leak inside an unpressurized cabin. What does the altimeter do? ●
- It slightly increases the indicated altitude
 - It slightly decreases the indicated altitude
 - It holds the altitude it had when the leakage started
- 321.** An aircraft cruises at constant altitude and constant airspeed with a blocked static system and then begins a climb. What does the altimeter do? ●
- It begins to increase its indication as the climb starts
 - It continues to display the same altitude
 - It jumps once and then follows the climb correctly
- 322.** The reason why the airspeed indication decreases during a climb with a blocked static system ●
- The total pressure and the static pressure both decrease together
 - The total pressure from the pitot tube stays constant while the static pressure decreases
 - The total pressure from the pitot tube decreases while the trapped static pressure stays constant
- 323.** The airspeed indication with a blocked static system during a climb compared with a descent ●
- Decreasing during both the climb and the descent
 - Decreasing during the climb and increasing during the descent
 - Increasing during the climb and decreasing during the descent
- 324.** The reason why the airspeed indication increases during a descent with a blocked static system ●
- The total pressure and the static pressure both increase together
 - The trapped static pressure increases while the total pressure remains constant
 - The total pressure from the pitot tube increases while the trapped static pressure remains constant
- 325.** Two aircraft climb, one with a blocked static system and one with a completely blocked pitot tube. What does the airspeed indicator show in each? ●
- A decreasing indication in both aircraft
 - A decreasing indication with the blocked static system and an increasing indication with the blocked pitot tube
 - An increasing indication with the blocked static system and a decreasing indication with the blocked pitot tube
- 326.** The airspeed indication during a descent with a blocked static system compared with a completely blocked pitot tube ●
- Decreasing with the blocked static system and increasing with the blocked pitot tube
 - Increasing with the blocked static system and decreasing with the blocked pitot tube
 - Increasing with both blockages
- 327.** The pitot tube becomes totally blocked in cruise and the thrust is then increased to accelerate. What does the airspeed indicator show? ●
- A rise in the indicated speed
 - A fall of the indicated speed to zero
 - No change in the indicated speed
- 328.** The pitot inlet of an aircraft is found blocked by ice. Which indication shows that the water drain hole is still open? ●
- The airspeed indication increases into the overspeed region
 - The airspeed indication decreases to zero
 - The airspeed indication stops responding to changes of thrust

- 329.** The pitot pressure and the airspeed indication when the pitot inlet is iced over but the drain is free ●
- The pitot pressure falls to static pressure and the airspeed indication decreases to zero
 - The pitot pressure remains constant and the airspeed indication decreases to zero
 - The pitot pressure falls to static pressure and the airspeed indication increases
- 330.** Two pitot probes are examined, one iced at the inlet only and one iced at the inlet and the drain. What does each condition say about the heating? ●
- Defective or deactivated heating for the inlet only, and insufficient heating when both are iced
 - Insufficient heating for the inlet only, and defective or deactivated heating when both are iced
 - Insufficient heating in both conditions
- 331.** The pitot inlet is iced over but the drain is free. What does this tell you about the heating? ●
- The heating is defective or deactivated
 - The heating is working correctly
 - The heating is insufficient
- 332.** Both the pitot inlet and the drain are iced over. What does this say about the heating and the pitot pressure? ●
- The heating is defective or deactivated and the pitot pressure remains constant
 - The heating is insufficient and the pitot pressure falls to static pressure
 - The heating is defective or deactivated and the pitot pressure falls to static pressure

Answers — Pitot static blockages and icing

- 292. c** Only the airspeed indicator is fed from the pitot system. (L1, PDF p. 431)
- 293. b** With static pressure trapped, neither instrument can register the climb. (L1, PDF p. 435)
- 294. b** The trapped static pressure never changes, so the VSI sees no rate at all. (L1, PDF p. 437)
- 295. b** Complete blockage covers both the inlet and the drain hole. (L1, PDF p. 439)
- 296. b** Those two run on static pressure, which the pitot blockage does not touch. (L1, PDF p. 439)
- 297. a** With pitot pressure equal to static pressure the airspeed indicator reads zero. (L1, PDF p. 441)
- 298. b** Fluctuating indications are the sign of water lying in the pipes. (L1, PDF p. 443)
- 299. a** With the pitot line open to the cabin the airspeed indication falls away to zero. (L2, PDF p. 431)
- 300. a** Both lines fill with the same cabin pressure, so the difference the indicator works on shrinks. (L2, PDF p. 433)
- 301. b** With both lines open to the same cabin pressure the difference shrinks and the indication falls. (L2, PDF p. 433)
- 302. b** Those two instruments see only static pressure, so the static leak result still applies. (L2, PDF p. 433)
- 303. b** Nothing changes while the trapped static pressure still matches the real one. (L2, PDF p. 435)
- 304. c** In the climb the falling total pressure works against a trapped static pressure, so the reading drops. (L2, PDF p. 435)
- 305. a** PDF page 440 prints the frozen altitude and zero vertical speed for the climb and this page the same pair for the descent, since both instruments work from the trapped static pressure. (L2, PDF p. 437)
- 306. b** In the descent the rising total pressure works against a trapped static pressure, so the reading climbs. (L2, PDF p. 437)
- 307. c** With the total pressure held fixed the indication is driven only by static pressure, which is what an altimeter responds to. (L2, PDF p. 439)
- 308. a** With trapped total pressure and falling static pressure the indication runs up. (L2, PDF p. 439)
- 309. b** Its reading now follows altitude change instead of speed. (L2, PDF p. 439)
- 310. a** The rising static pressure against a trapped total pressure drives the reading down. (L2, PDF p. 439)
- 311. c** The open drain lets the trapped pressure bleed down to static. (L2, PDF p. 441)
- 312. c** Both instruments are fed from the static system alone. (L2, PDF p. 441)
- 313. a** The pipes are not drained routinely, so water can collect in them. (L2, PDF p. 443)
- 314. a** Warming the trapped air in a sealed static system raises its pressure. (L2, PDF p. 443)

- 315. a** With the total pressure held fixed, only the static pressure varies, so ram pressure follows altitude. (L2, PDF p. 443)
- 316. c** With no cabin pressure the two pressures equalise and the indication falls to zero, while cabin pressure makes the result depend on altitude and cabin pressure. (L3, PDF p. 431)
- 317. b** The airspeed indicator works on the difference between the two pressures, which is now almost nil. (L3, PDF p. 431)
- 318. b** With a pressurized cabin the two varying pressures make the indication unpredictable. (L3, PDF p. 431)
- 319. a** PDF page 436 prints that a pitot leak affects the airspeed indicator only, and this page adds the altimeter and the vertical speed indication once the static system leaks as well. (L3, PDF p. 433)
- 320. a** This page refers the altimeter back to the static leakage result, and PDF page 432 prints that result for an unpressurized cabin as a slightly increased indicated altitude. (L3, PDF p. 433)
- 321. b** The trapped static pressure cannot change, so the altimeter holds the same altitude in level flight and through the climb. (L3, PDF p. 435)
- 322. c** Only the total pressure falls; the blocked static side cannot follow it. (L3, PDF p. 435)
- 323. b** PDF page 440 prints the decreasing indication in the climb and this page the increasing one in the descent, because the trapped static pressure stays fixed while the total pressure falls and then rises. (L3, PDF p. 437)
- 324. c** Only the total pressure rises; the blocked static side stays where it was. (L3, PDF p. 437)
- 325. b** PDF page 440 prints the decreasing indication with the static side blocked, while this page prints the increasing one with the pitot side blocked, because the falling static pressure then drives the instrument. (L3, PDF p. 439)
- 326. b** PDF page 442 prints the increasing indication with the static side blocked, while this page prints the decreasing one with the pitot side blocked, because the rising static pressure then drives the instrument. (L3, PDF p. 439)
- 327. c** At constant altitude neither pressure changes, so the reading is frozen whatever the thrust. (L3, PDF p. 439)
- 328. b** With the drain open the pitot pressure falls to static and the indication drops to zero, which a completely blocked tube does not do. (L3, PDF p. 441)
- 329. a** The open drain lets the pitot pressure fall to static pressure, leaving no difference for the indicator to show. (L3, PDF p. 443)
- 330. b** Ice at the inlet alone points to heating that is too weak, while ice at both openings points to heating that is defective or switched off. (L3, PDF p. 443)
- 331. c** Ice only at the inlet points to heating that is working but not enough. (L3, PDF p. 443)
- 332. a** Complete icing means the heating has failed or is off, and the sealed pitot pressure cannot change. (L3, PDF p. 443)